User Manual

for S32K3 ETH Driver

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Chapter 1

Revision History

Revision	Date	Author	Description
1.0	31.03.2022	NXP RTD Team	Prepared for release RTD S32K3 2.0.0

Chapter 2

Introduction

- Supported Derivatives
- Overview
- About This Manual
- Acronyms and Definitions
- Reference List

This User Manual describes NXP Semiconductors' AUTOSAR Ethernet Driver for S32K3.

AUTOSAR Ethernet Driver configuration parameters description can be found in the Tresos Configuration Plugin section. Deviations from the specification are described in the Deviations from Requirements section.

AUTOSAR Ethernet driver requirements and APIs are described in the Ethernet Driver Software Specification Document (version 4.4.0).

2.1 Supported Derivatives

The software described in this document is intended to be used with the following microcontroller devices of NXP Semiconductors:

- s32k344_mapbga257
- s32k344_mqfp172
- s32k324_mapbga257
- s32k324_mqfp172
- s32k314_mapbga257
- s32k314_mqfp172
- s32k312_mqfp172
- s32k312_mqfp100

- s32k322_mqfp172
- s32k322_mqfp100
- s32k342_mqfp172
- s32k342_mqfp100
- s32k341_mqfp172
- s32k341_mqfp100

All of the above microcontroller devices are collectively named as S32K3.

2.2 Overview

AUTOSAR (AUTomotive Open System ARchitecture) is an industry partnership working to establish standards for software interfaces and software modules for automobile electronic control systems.

AUTOSAR:

- paves the way for innovative electronic systems that further improve performance, safety and environmental friendliness.
- is a strong global partnership that creates one common standard: "Cooperate on standards, compete on implementation".
- is a key enabling technology to manage the growing electrics/electronics complexity. It aims to be prepared for the upcoming technologies and to improve cost-efficiency without making any compromise with respect to quality.
- facilitates the exchange and update of software and hardware over the service life of the vehicle.

2.3 About This Manual

This Technical Reference employs the following typographical conventions:

- **Boldface** style: Used for important terms, notes and warnings.
- *Italic* style: Used for code snippets in the text. Note that C language modifiers such "const" or "volatile" are sometimes omitted to improve readability of the presented code.

Notes and warnings are shown as below:

Note

This is a note.

Warning

This is a warning

2.4 Acronyms and Definitions

Term	Definition
API	Application Programming Interface
AUTOSAR	Automotive Open System Architecture
DEM	Diagnostic Event Manager
DET	Default Error Tracer
ETH	Ethernet
ETHIF	Ethernet Interface
ETHTRCV	Ethernet Transceiver
ETHSWT	Ethernet Switch
MCU	Micro controller Unit
MII	Media Independent Interface
N/A	Not Available
RMII	Reduced Media Independent Interface
RAM	Random Access Memory

- The term "Ethernet Controller" is related to the hardware module providing the Ethernet functionality.
- The term "Ethernet Driver" is related to the software handling the Ethernet Controller.
- The term "Application" is used for the software utilizing the Ethernet Driver.

Term	Definition
API	Application Programming Interface
AUTOSAR	Automotive Open System Architecture
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- The term "Application" is used for the software utilizing the Ethernet Driver.

2.5 Reference List

#	Title	Version
1	Specification of ETH Driver	AUTOSAR Release 4.4.0
2	S32K3XX Reference Manual	S32K3xx Reference Manual, Rev. 3, 10/2021
3	Datasheet	S32K3xx Data Sheet, Rev. 3 10/2021
4	Errata	S32K312, Mask Set Errata for Mask 0P09C, Rev. 10/Jan/2022,
		S32K342: Mask Set Errata for Mask 0P97C, Rev. 13 Dec 2021,
		S32K344: Mask Set Errata for Mask 0P55A/1P55A, Rev. 30/Sep/2021.

Chapter 3

Driver

- Requirements
- Driver Design Summary
- Hardware Resources
- Deviations from Requirements
- Driver Limitations
- Driver usage and configuration tips
- Runtime errors
- Symbolic Names Disclaimer

3.1 Requirements

Requirements for this driver are detailed in the AUTOSAR 4.4.0 Ethernet Driver Software Specification document (See Table Reference List)

3.2 Driver Design Summary

The Ethernet Driver controls the Ethernet Media Access Controller (EMAC) module of the S32K3XX device. It provides the following features:

- Configuration and initialization of the Ethernet Controller
- Switching the Ethernet Controller on and off
- Reception and transmission of Ethernet frames
- Access to the Ethernet Controller's RMON counters (RxStats, TxStats and TxErrorCounterValues)
- Access to the Ethernet Transceiver device registers through MDIO
- Handling of the Ethernet Controller's interrupt requests

- Half and full duplex operation support
- Various PHY Interfaces (10/100 Mbps on MII and RMII)
- Time Synchronization over Ethernet (The gPTP stack has to be provided by the upper layers)
- Time Aware shaper, frame preemption for time sensitive networking.
- Checksum hardware offloading for IPv4/IPv6 with TCP, UDP or ICMP

3.3 Hardware Resources

For the Ethernet driver, there are no differences in hardware resources between derivatives.

The hardware module configured by the Ethernet driver is EMAC (Ethernet Media Access Controller). It has a single hardware instance (EMAC_0).

The Ethernet controller to microcontroller pin mapping can be done using the file **S32K3XX_IOMUX.xlsx** attached to the Reference Manual.

3.4 Deviations from Requirements

The driver deviates from the AUTOSAR ETH Driver software specification in some places. The table below identifies the AUTOSAR requirements that are not implemented or out of scope for the ETH Driver.

Term	Definition
N/S	Out of scope
N/I	Not implemented
N/F	Not fully implemented

Below table identifies the AUTOSAR requirements that are not fully implemented, implemented differently or out of scope for the ETH driver.

Requirement	Status	Description	Notes
ECUC_Eth_00035	N/S	Name - EthGetDropCountApi - Parent Container - EthGeneral - Description - Enables / Disables Eth_Get↩ CounterValues API. - Multiplicity - 1 - Type - EcucBooleanParamDef - Default value - – - Post-Build Variant Value - false - Value Configuration Class - Pre-compile time - X - All Variants - Link time - – - - Post-build time - – - - Scope / Dependency - scope: local -	Just exist in ASR4.2.X

Requirement	Status	Description	Notes
ECUC_Eth_00036	N/S	Name - EthGetEtherStatsApi - Parent Container - EthGeneral - Description - Enables / Disables Eth_Get↩ EtherStats API. - Multiplicity - 1 - Type - EcucBooleanParamDef - Default value - – - Post-Build Variant Value - false - Value Configuration Class - Pre-compile time - X - All Variants - Link time - – - - Post-build time - – - - Scope / Dependency - scope: local -	Just exist in ASR4.2.X
SWS_Eth_00257	N/F	DRAFT: The Ethernet Driver module shall reject configurations with partition mappings which are not supported by the implementation.	Derivative supports multicore: S32↩ K324 Derivatives support only one core: S32K344, S32K314
SWS_Eth_00258	N/F	DRAFT: If the driver manages several Ethernet controllers and if a subset of these controllers share peripheral resources or are somehow coupled (E.↩ g. Communication control can only be done globally for all controllers), Ethernet driver shall emulate independent controllers to the upper layers. The coordination (E.g. Communication control) has to be done by the upper layer modules.	Derivative supports multicore: S32↩ K324 Derivatives support only one core: S32K344, S32K314
SWS_Eth_00260	N/F	DRAFT: The ECUC partitions referenced by EthCtrlEcucPartitionRef shall be a subset of the ECUC partitions referenced by EthEcuc↩ PartitionRef.	Derivative supports multicore: S32↩ K324 Derivatives support only one core: S32K344, S32K314
SWS_Eth_00261	N/F	DRAFT: EthCtrlConfig, EthTrcv↩ Config and EthSwtConfig (if existent in configuration) of one communication channel shall all reference the same ECUC partition	Derivative supports multicore: S32↩ K324 Derivatives support only one core: S32K344, S32K314
SWS_Eth_00259	N/F	DRAFT: The module will operate as an independent instance in each of the partitions, means the called API will only target the partition it is called in.	Derivative supports multicore: S32↩ K324 Derivatives support only one core: S32K344, S32K314

Requirement	Status	Description	Notes
ECUC_Eth_00065	N/F	Name - EthCtrlEcucPartitionRef - Parent Container - EthCtrlConfig - Description - Maps the Ethernet controller to zero or one ECUC partitions. The ECUC partition referenced is a subset of the ECUC partitions where the Ethernet driver is mapped to.↩ Tags: atp.Status=draft - Multiplicity - 0..1 - Type - Reference to [Ecuc↩ Partition] - Post-Build Variant Multiplicity - true - Post-Build Variant Value - true - Multiplicity Configuration Class - Pre-compile time - X - All Variants - Link time - – - - Post-build time - – - - Value Configuration Class - Pre-compile time - X - All Variants - Link time - – - - Post-build time - – - - Scope / Dependency - scope: ECU -	Derivative supports multicore: S32↩ K324 Derivatives support only one core: S32K344, S32K314
ECUC_Eth_00064	N/F	Name - EthEcucPartitionRef - Parent Container - EthGeneral - Description - Maps the Ethernet driver to zero or multiple ECUC partitions to make the modules API available in this partition. The Ethernet driver will operate as an independent instance in each of the partitions.Tags: atp.Status=draft - Multiplicity - 0..∗ - Type - Reference to [EcucPartition] - Post-Build Variant Multiplicity - true - Post-Build Variant Value - true - Multiplicity Configuration Class - Pre-compile time - X - All Variants - Link time - – - - Post-build time - – - - Value Configuration Class - Pre-compile time - X - All Variants - Link time - – - - Post-build time - – - - Scope / Dependency - scope: ECU -	Derivative supports multicore: S32↩ K324 Derivatives support only one core: S32K344, S32K314

3.5 Driver Limitations

The Ethernet Driver has the following limitations:

- The API functions *Eth_ReadMii* and *Eth_WriteMii* (and their counterparts *Eth_ReadMmd* and *Eth_Write↩ Mmd*) are synchronous instead of asynchronous due to a variety of reasons further detailed in `this` AUTOSAR ticket.
- The length of a single received frame (including the 14-bytes Ethernet frame header and the 4-bytes FCS) must be less than or equal to EthCtrlConfigIngressFifoBufLenByte.

- When preemption enabled, MMC can incorrectly operate for Tx when there are so many intermediate fragment frames that SMD-Sx + MCRC_0 + SMD_Cx_1 + FragCount_1 + MCRC_1 + ... + SMD_Cx_n + Frag↩ Count_n + FCS would make the frame transmission time to exceed the jabber limit (i.e. the time limit to transmit an uninterrupted/unfragmented frame with the maximum configured MTU). This is due to Errata ERR050597 can be find in Errata documentation in <Reference List section>.

3.6 Driver usage and configuration tips

None

3.7 Runtime errors

The Ethernet driver generates the following DEM extended production errors at runtime:

3.7.1 Standardized Extended Production Errors

Function	Error code	Condition triggering the error
Eth_MainFunction	ETH_E_ACCESS	Controller access failed
Eth_MainFunction	ETH_E_RX_FRAMES_LOST	Rx frame lost detected
Eth_MainFunction	ETH_E_CRC	CRC failure detected
Eth_MainFunction	ETH_E_UNDERSIZEFRAME	Undersized frame detected
Eth_MainFunction	ETH_E_OVERSIZEFRAME	Oversized frame detected
Eth_MainFunction	ETH_E_ALIGNMENT	Alignment error detected
Eth_MainFunction	ETH_E_SINGLECOLLISION	Single collision detected
Eth_MainFunction	ETH_E_MULTIPLECOLLISION	Multiple collision detected
Eth_MainFunction	ETH_E_LATECOLLISION	Late collision detected

Function	Error code	Condition triggering the error
Eth_Ipw_SafetyIrqCallback	ETH_E_SAFETY	ECC error, datapath parity error, FSM state parity error, FSM timeout error

3.7.2 Vendor-Specific Extended Production Errors

3.8 Symbolic Names Disclaimer

All containers having symbolicNameValue set to TRUE in the AUTOSAR schema will generate defines like:

#define <Mip>Conf_<Container_ShortName>_<Container_ID>

For this reason it is forbidden to duplicate the names of such containers across the RTD configurations or to use names that may trigger other compile issues (e.g. match existing `#ifdefs` arguments).

Chapter 4

Tresos Configuration Plug-in

This chapter describes the Tresos configuration plug-in for the driver. All the parameters are described below.

- Module Eth
  - Container EthGeneral
    - Parameter EthDevErrorDetect
    - Parameter EthGetDropCountApi
    - Parameter EthGetEtherStatsApi
    - Parameter EthGetCounterValuesApi
    - Parameter EthGetRxStatsApi
    - Parameter EthGetTxStatsApi
    - Parameter EthGetTxErrorCounterValuesApi
    - Parameter EthGlobalTimeSupport
    - Parameter EthVersionInfoApi
    - Parameter EthIndex
    - Parameter EthMainFunctionPeriod
    - Parameter EthMaxCtrlsSupported
    - Reference EthEcucPartitionRef
    - Container EthCtrlOffloading
      - Parameter EthCtrlEnableOffloadChecksumICMP
      - Parameter EthCtrlEnableOffloadChecksumIPv4
      - Parameter EthCtrlEnableOffloadChecksumTCP
      - Parameter EthCtrlEnableOffloadChecksumUDP
    - Container EthGeneralVendorSpecific
      - Parameter EthDisableDemEventDetect
      - Parameter EthEnableUserModeSupport
      - Parameter EthMulticoreSupport
      - Parameter EthUpdatePhysAddrFilterApi
      - Parameter EthSwtManagementSupportApi
      - Parameter EthTimeoutMethod
      - Parameter EthTimeoutDuration
  - Container EthConfigSet
    - Container EthCtrlConfig

- Parameter EthCtrlEnableMii
- Parameter EthCtrlEnableMmd
- Parameter EthCtrlEnableRxInterrupt
- Parameter EthCtrlEnableTxInterrupt
- Parameter EthCtrlIdx
- Parameter EthCtrlMacLayerType
- Parameter EthCtrlMacLayerSubType
- Parameter EthCtrlMacLayerSpeed
- Parameter EthCtrlPhyAddress
- Reference EthCtrlEcucPartitionRef
- Container EthCtrlVendorSpecific
- Container EthCtrlConfigGeneral
- Parameter EthCtrlEnableAtInit
- Parameter EthCtrlAllocateTxDataBuffers
- Parameter EthCtrlAllocateRxDataBuffers
- Parameter EthEthTrcvDriverVendorIdAndApiInfix
- Parameter EthEthSwtDriverVendorIdAndApiInfix
- Parameter EthDuplexMode
- Parameter EthTimeStampRequiredAccuracy
- Reference EthModuleReferenceClock
- Reference EthTimeStampReferenceClock
- Container EthCtrlConfigSafety
- Parameter EthCtrlSafetyCallback
- Parameter EthCtrlEnableEcc
- Parameter EthCtrlEnableDpp
- Parameter EthCtrlEnableFsmSpp
- Parameter EthCtrlEnableFsmTimeout
- Reference ETH_E_SAFETY
- Container EthCtrlConfigMac
- Parameter MAC_CONFIG_CRC_STRIPPING
- Parameter MAC_CONFIG_AUTO_PAD
- Parameter MAC_CONFIG_JUMBO_PKT_EN
- Parameter MAC_CONFIG_LOOPBACK
- Parameter MAC_CONFIG_ENABLE_CRS_FD
- Parameter MAC_CONFIG_DISABLE_RECEIVE_OWN_HD
- Parameter MAC_CONFIG_DISABLE_CRS_HD
- Parameter MAC_CONFIG_DISABLE_RETRY
- Parameter MAC_CONFIG_DEFERRAL_CHECK_HD
- Container EthCtrlConfigPacketFilter
- Parameter PKT_FILTER_RECV_ALL
- Parameter PKT_FILTER_HASH_OR_PERFECT_FILTER
- Parameter PKT_FILTER_BLOCK_CONTROL_PKTS
- Parameter PKT_FILTER_BLOCK_PAUSE_PKTS
- Parameter PKT_FILTER_PASS_ALL_CONTROL_PKTS
- Parameter PKT_FILTER_PASS_CONTROL_PKTS_ADDR_MATCH
- Parameter PKT_FILTER_DISABLE_BROADCAST
- Parameter PKT_FILTER_PASS_ALL_MULTICAST

- Parameter PKT_FILTER_DST_ADDR_INV_FILTER_EN
- Parameter PKT_FILTER_HASH_MULTICAST
- Parameter PKT_FILTER_HASH_UNICAST
- Parameter PKT_FILTER_PROMISCUOUS_MODE
- Container EthCtrlConfigEgress
- Reference EthCtrlConfigEgressLastSchedulerRef
- Container EthCtrlConfigEgressFifo
- Parameter EthCtrlConfigEgressFifoBufLenByte
- Parameter EthCtrlConfigEgressFifoBufTotal
- Parameter EthCtrlConfigEgressFifoIdx
- Parameter EthCtrlConfigEgressFifoCallback
- Parameter EthCtrlConfigEgressFifoPriorityAssignment
- Container EthCtrlConfigScheduler
- Container EthCtrlConfigSchedulerPredecessor
- Parameter EthCtrlConfigSchedulerPredecessorOrder
- Reference EthCtrlConfigSchedulerPredecessorRef
- Container EthCtrlConfigSchedulerPredecessorVendorSpecific
- Parameter EthQueueBandwidth
- Container EthCtrlConfigSchedulerVendorSpecific
- Parameter EthTxSchedulerAlgorithm
- Parameter EthTotalPacketsPerCycle
- Parameter EthTotalQuantumPerCycle
- Container EthCtrlConfigShaper
- Parameter EthCtrlConfigShaperIdleSlope
- Reference EthCtrlConfigShaperPredecessorFifoRef
- Container EthCtrlConfigShaperVendorSpecific
- Parameter EthCtrlConfigHiCredit
- Parameter EthCtrlConfigLoCredit
- Container EthCtrlConfigTimeAwareShaper
- Container EthCtrlConfigTimeAwareShaper
- Parameter EthGateConfigBaseTimeInSecond
- Parameter EthGateConfigBaseTimeInNanoSecond
- Parameter EthGateConfigCycleTime
- Parameter EthGateConfigExtendTime
- Parameter EthCtrlEnablePreemtion
- Parameter EthReleaseAdvanceTime
- Parameter EthHoldAdvanceTime
- Container EthGateControlList
- Parameter EthCtrlConfigInterval
- Container EthGateControlFifo
- Parameter EthCtrlConfigGateStatus
- Reference EthCtrlConfigFifoRef
- Container EthPreemptionClassification
- Parameter EthCtrlConfigPreemptionClassification
- Reference EthPreemptionFifoRef
- Container EthCtrlConfigIngress
- Container EthCtrlConfigIngressFifo

- Parameter EthCtrlConfigIngressFifoBufLenByte
- Parameter EthCtrlConfigIngressFifoBufTotal
- Parameter EthCtrlConfigIngressFifoIdx
- Parameter EthCtrlConfigIngressFifoCallback
- Parameter EthCtrlConfigIngressFifoPriorityAssignment
- Container EthDemEventParameterRefs
- Reference ETH_E_ACCESS
- Reference ETH_E_RX_FRAMES_LOST
- Reference ETH_E_CRC
- Reference ETH_E_UNDERSIZEFRAME
- Reference ETH_E_OVERSIZEFRAME
- Reference ETH_E_ALIGNMENT
- Reference ETH_E_SINGLECOLLISION
- Reference ETH_E_MULTIPLECOLLISION
- Reference ETH_E_LATECOLLISION

– Container CommonPublishedInformation

* Parameter ModuleId
* Parameter VendorId
* Parameter VendorApiInfix
* Parameter ArReleaseMajorVersion
* Parameter ArReleaseMinorVersion
* Parameter ArReleaseRevisionVersion
* Parameter SwMajorVersion
* Parameter SwMinorVersion
* Parameter SwPatchVersion

4.1 Module Eth

Configuration of the Eth (Ethernet Driver) module.

Included containers:

- EthGeneral
- EthConfigSet
- CommonPublishedInformation

Property	Value
type	ECUC-MODULE-DEF
lowerMultiplicity	0
upperMultiplicity	1
postBuildVariantSupport	true
supportedConfigVariants	VARIANT-PRE-COMPILE, VARIANT-LINK-TIME, VARIANT-POST-BUILD

4.2 Container EthGeneral

General configuration of Ethernet Driver module.

Included subcontainers:

- EthCtrlOffloading
- EthGeneralVendorSpecific

Property	Value
type	ECUC-PARAM-CONF-CONTAINER-DEF
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A

4.3 Parameter EthDevErrorDetect

Enables / Disables development error detection.

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	AUTOSAR_ECUC
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-LINK-TIME: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE
defaultValue	true

4.4 Parameter EthGetDropCountApi

Obsolete and kept only for backwards compatibility. The Eth_GetDropCount API has been replaced with Eth_GetCounterVal since R4.3.1. Use parameter EthGetCounterValuesApi instead.

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	AUTOSAR_ECUC
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-LINK-TIME: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE
defaultValue	false

4.5 Parameter EthGetEtherStatsApi

Obsolete and kept only for backwards compatibility. The Eth_GetEtherStats API has been replaced with the Eth_GetTxStats and Eth_GetRxStats APIs since R4.3.1.

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	AUTOSAR_ECUC
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-LINK-TIME: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE
defaultValue	false

4.6 Parameter EthGetCounterValuesApi

Enables / Disables Eth_GetCounterValues API.

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	NXP

Property	Value
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-LINK-TIME: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE
defaultValue	false

4.7 Parameter EthGetRxStatsApi

Enables / Disables Eth_GetRxStats API.

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-LINK-TIME: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE
defaultValue	false

4.8 Parameter EthGetTxStatsApi

Enables / Disables Eth_GetTxStats API.

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	AUTOSAR_ECUC
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1

Property	Value
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-LINK-TIME: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE
defaultValue	false

4.9 Parameter EthGetTxErrorCounterValuesApi

Enables / Disables Eth_GetTxErrorCounterValues API.

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	AUTOSAR_ECUC
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-LINK-TIME: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE
defaultValue	false

4.10 Parameter EthGlobalTimeSupport

Enables/Disables the GlobalTime APIs used amongst others by Global Time Synchronization over Ethernet.

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	AUTOSAR_ECUC
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false

Property	Value
valueConfigClasses	VARIANT-LINK-TIME: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE
defaultValue	false

4.11 Parameter EthVersionInfoApi

Enables / Disables Eth_GetVersionInfo API.

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	AUTOSAR_ECUC
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-LINK-TIME: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE
defaultValue	true

4.12 Parameter EthIndex

Specifies the InstanceId of this module instance. If only one instance is present it shall have the Id 0.

Property	Value
type	ECUC-INTEGER-PARAM-DEF
origin	AUTOSAR_ECUC
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-LINK-TIME: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE

Property	Value
defaultValue	0
max	255
min	0

4.13 Parameter EthMainFunctionPeriod

The period between successive calls to the main function in seconds.

Ethernet driver does not require this information but the BSW scheduler.

Property	Value
type	ECUC-FLOAT-PARAM-DEF
origin	AUTOSAR_ECUC
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-LINK-TIME: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE
defaultValue	0.001
max	65.535
min	0.001

4.14 Parameter EthMaxCtrlsSupported

Limits the total number of supported controllers.

Property	Value
type	ECUC-INTEGER-PARAM-DEF
origin	AUTOSAR_ECUC
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
	VARIANT-LINK-TIME: PRE-COMPILE

valueConfigClasses

Property	Value
	VARIANT-POST-BUILD: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE
defaultValue	1
max	1
min	1

4.15 Reference EthEcucPartitionRef

Maps the Ethernet driver to zero or multiple ECUC partitions to make the module's API available in this partition.

The Ethernet driver will operate as an independent instance in each of the partitions.

Property	Value
type	ECUC-REFERENCE-DEF
origin	AUTOSAR_ECUC
lowerMultiplicity	0
upperMultiplicity	Infinite
postBuildVariantMultiplicity	true
multiplicityConfigClasses	VARIANT-POST-BUILD: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-LINK-TIME: PRE-COMPILE
postBuildVariantValue	true
valueConfigClasses	VARIANT-POST-BUILD: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-LINK-TIME: PRE-COMPILE
requiresSymbolicNameValue	False
destination	/AUTOSAR/EcucDefs/EcuC/EcucPartitionCollection/EcucPartition

4.16 Container EthCtrlOffloading

Configuration of hardware offloading features.

Included subcontainers:

- None

Property	Value
type	ECUC-PARAM-CONF-CONTAINER-DEF
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A

4.17 Parameter EthCtrlEnableOffloadChecksumICMP

Enables / Disables hardware offloading for ICMP checksums.

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	AUTOSAR_ECUC
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-LINK-TIME: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE
defaultValue	false

4.18 Parameter EthCtrlEnableOffloadChecksumIPv4

Enables / Disables hardware offloading for IPv4 checksums.

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	AUTOSAR_ECUC
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-LINK-TIME: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE
defaultValue	false

4.19 Parameter EthCtrlEnableOffloadChecksumTCP

Enables / Disables hardware offloading for TCP checksums.

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	AUTOSAR_ECUC
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-LINK-TIME: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE
defaultValue	false

4.20 Parameter EthCtrlEnableOffloadChecksumUDP

Enables / Disables hardware offloading for UDP checksums.

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	AUTOSAR_ECUC
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-LINK-TIME: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE
defaultValue	false

4.21 Container EthGeneralVendorSpecific

General vendor-specific configuration of Ethernet Driver.

Included subcontainers:

- None

Property	Value
type	ECUC-PARAM-CONF-CONTAINER-DEF
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A

4.22 Parameter EthDisableDemEventDetect

When this option is enabled, no DEM events are reported.

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-LINK-TIME: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE
defaultValue	false

4.23 Parameter EthEnableUserModeSupport

When this parameter is enabled, the Eth module will adapt to run from User Mode, with the following measures:

(if applicable) a) configuring REG_PROT for the Eth Controllers so that the registers under protection eth be accessed from user mode by setting UAA bit in REG_PROT_GCR to 1.

(if applicable) b) using 'call trusted function' stubs for all internal function calls that access registers requiring supervisor mode.

(if applicable) c) other module specific measures for more information, please see chapter 5.7 User Mode Support in IM

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	NXP
symbolicNameValue	false

Property	Value
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-LINK-TIME: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE
defaultValue	false

4.24 Parameter EthMulticoreSupport

Switches multicore support on or off:

False: For all variants, no EcucPartition shall be referenced in EthEcucPartitionRef.

True: For all variants, at least one EcucPartition needs to be referenced in EthEcucPartitionRef.

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-LINK-TIME: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE
defaultValue	true

4.25 Parameter EthUpdatePhysAddrFilterApi

Enables/Disables Eth_UpdatePhysAddrFilter API.

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	NXP
symbolicNameValue	false

Property	Value
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-LINK-TIME: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE
defaultValue	false

4.26 Parameter EthSwtManagementSupportApi

Enables / Disables the Switch Management APIs.

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-LINK-TIME: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE
defaultValue	false

4.27 Parameter EthTimeoutMethod

Configures the timeout method.

Based on this selection a certain timeout method from OsIf will be used in the driver.

Note: If SystemTimer or CustomTimer are selected make sure the corresponding timer is enabled in OsIf General configuration.

Note: Implementation Specific Parameter.

Property	Value
type	ECUC-ENUMERATION-PARAM-DEF

Property	Value
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-LINK-TIME: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
defaultValue	OSIF_COUNTER_DUMMY
literals	['OSIF_COUNTER_SYSTEM', 'OSIF_COUNTER_CUSTOM', 'OSIF_CO↩ UNTER_DUMMY']

4.28 Parameter EthTimeoutDuration

The unit of measurement is given in number of microseconds.

Property	Value
type	ECUC-INTEGER-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-LINK-TIME: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE
defaultValue	1000
max	65535
min	1

4.29 Container EthConfigSet

This container contains the configuration parameters and sub containers of the AUTOSAR Eth module.

Included subcontainers:

- EthCtrlConfig

Property	Value
type	ECUC-PARAM-CONF-CONTAINER-DEF
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A

4.30 Container EthCtrlConfig

Configuration of the individual controller

Included subcontainers:

- EthCtrlVendorSpecific
- EthCtrlConfigEgress
- EthCtrlConfigTimeAwareShaper
- EthCtrlConfigIngress
- EthDemEventParameterRefs

Property	Value
type	ECUC-PARAM-CONF-CONTAINER-DEF
lowerMultiplicity	1
upperMultiplicity	255
postBuildVariantMultiplicity	true
multiplicityConfigClasses	VARIANT-LINK-TIME: LINK
	VARIANT-POST-BUILD: POST-BUILD
	VARIANT-PRE-COMPILE: PRE-COMPILE

4.31 Parameter EthCtrlEnableMii

Enables / Disables Media Independent Interface (MII) for transceiver access.

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	AUTOSAR_ECUC
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1

Property	Value
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-LINK-TIME: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE
defaultValue	false

4.32 Parameter EthCtrlEnableMmd

Enables / Disables Clause 45 Media Independent Interface (MII) for transceiver access.

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-LINK-TIME: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
defaultValue	false

4.33 Parameter EthCtrlEnableRxInterrupt

Enables / Disables receive interrupt.

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	AUTOSAR_ECUC
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false

Property	Value
valueConfigClasses	VARIANT-LINK-TIME: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE
defaultValue	true

4.34 Parameter EthCtrlEnableTxInterrupt

Enables / Disables transmit interrupt.

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	AUTOSAR_ECUC
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-LINK-TIME: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE
defaultValue	true

4.35 Parameter EthCtrlIdx

Specifies the instance ID of the configured controller.

Property	Value
type	ECUC-INTEGER-PARAM-DEF
origin	AUTOSAR_ECUC
symbolicNameValue	true
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-LINK-TIME: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE

Property	Value
defaultValue	0
max	0
min	0

4.36 Parameter EthCtrlMacLayerType

Defines the MAC layer type of the ethernet controller.

Property	Value
type	ECUC-ENUMERATION-PARAM-DEF
origin	AUTOSAR_ECUC
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-LINK-TIME: LINK
	VARIANT-POST-BUILD: POST-BUILD
	VARIANT-PRE-COMPILE: PRE-COMPILE
defaultValue	ETH_MAC_LAYER_TYPE_XMII
literals	['ETH_MAC_LAYER_TYPE_XMII', 'ETH_MAC_LAYER_TYPE_XGM↩ II', 'ETH_MAC_LAYER_TYPE_XXGMII']

4.37 Parameter EthCtrlMacLayerSubType

Defines the MAC layer subtype of the ethernet controller.

Property	Value
type	ECUC-ENUMERATION-PARAM-DEF
origin	AUTOSAR_ECUC
symbolicNameValue	false
lowerMultiplicity	0
upperMultiplicity	1
postBuildVariantMultiplicity	true
multiplicityConfigClasses	VARIANT-LINK-TIME: LINK
	VARIANT-POST-BUILD: LINK
	VARIANT-PRE-COMPILE: PRE-COMPILE
postBuildVariantValue	true
	VARIANT-LINK-TIME: LINK

valueConfigClasses

Property	Value
	VARIANT-POST-BUILD: LINK
	VARIANT-PRE-COMPILE: PRE-COMPILE
defaultValue	STANDARD
literals	['REDUCED', 'SERIAL', 'STANDARD']

4.38 Parameter EthCtrlMacLayerSpeed

Defines the baud rate of the MAC layer.

Property	Value
type	ECUC-ENUMERATION-PARAM-DEF
origin	AUTOSAR_ECUC
symbolicNameValue	false
lowerMultiplicity	0
upperMultiplicity	1
postBuildVariantMultiplicity	true
multiplicityConfigClasses	VARIANT-LINK-TIME: LINK
	VARIANT-POST-BUILD: LINK
	VARIANT-PRE-COMPILE: PRE-COMPILE
postBuildVariantValue	true
valueConfigClasses	VARIANT-LINK-TIME: LINK
	VARIANT-POST-BUILD: LINK
	VARIANT-PRE-COMPILE: PRE-COMPILE
defaultValue	ETH_MAC_LAYER_SPEED_10M
literals	['ETH_MAC_LAYER_SPEED_10M', 'ETH_MAC_LAYER_SPEED_100M', 'ETH_MAC_LAYER_SPEED_200M', 'ETH_MAC_LAYER_SPEED_1G', 'ETH_MAC_LAYER_SPEED_10G']

4.39 Parameter EthCtrlPhyAddress

Specifies the unique 48-bit physical address (MAC address) of the controller in network byte order.

Property	Value
type	ECUC-STRING-PARAM-DEF
origin	AUTOSAR_ECUC
symbolicNameValue	false
lowerMultiplicity	0
upperMultiplicity	1
postBuildVariantMultiplicity	true
	VARIANT-LINK-TIME: LINK

multiplicityConfigClasses

Property	Value
	VARIANT-POST-BUILD: POST-BUILD
	VARIANT-PRE-COMPILE: PRE-COMPILE
postBuildVariantValue	true
valueConfigClasses	VARIANT-LINK-TIME: LINK
	VARIANT-POST-BUILD: POST-BUILD
	VARIANT-PRE-COMPILE: PRE-COMPILE
defaultValue	66:55:44:33:22:11

4.40 Reference EthCtrlEcucPartitionRef

Maps the Ethernet controller to zero or one ECUC partition.

The ECUC partition referenced is a subset of the ECUC partitions where the Ethernet driver is mapped to.

Property	Value
type	ECUC-REFERENCE-DEF
origin	AUTOSAR_ECUC
lowerMultiplicity	0
upperMultiplicity	1
postBuildVariantMultiplicity	true
multiplicityConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-LINK-TIME: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
postBuildVariantValue	true
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-LINK-TIME: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
requiresSymbolicNameValue	False
destination	/AUTOSAR/EcucDefs/EcuC/EcucPartitionCollection/EcucPartition

4.41 Container EthCtrlVendorSpecific

Vendor specific configuration of the individual controller

Included subcontainers:

- EthCtrlConfigGeneral
- EthCtrlConfigSafety
- EthCtrlConfigMac
- EthCtrlConfigPacketFilter

Property	Value
type	ECUC-PARAM-CONF-CONTAINER-DEF
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A

4.42 Container EthCtrlConfigGeneral

General Controller Configuration.

Included subcontainers:

- None

Property	Value
type	ECUC-PARAM-CONF-CONTAINER-DEF
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A

4.43 Parameter EthCtrlEnableAtInit

Enabled - The controller is enabled at initialization.

Disabled - The controller is left disabled at initialization and must be later enabled by calling Gmac_Ip_EnableController.

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-LINK-TIME: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE
defaultValue	false

4.44 Parameter EthCtrlAllocateTxDataBuffers

Enabled - The driver will allocate both Tx Buffer Descriptors and Tx Data Buffers.

Disabled - The driver will only allocate Tx Buffer Descriptors. The Tx Data Buffers will be allocated by the application.

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-LINK-TIME: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE
defaultValue	true

4.45 Parameter EthCtrlAllocateRxDataBuffers

Enabled - The driver will allocate both Rx Buffer Descriptors and Rx Data Buffers.

Disabled - The driver will only allocate Rx Buffer Descriptors. The Rx Data Buffers will be allocated by the application.

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-LINK-TIME: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE
defaultValue	true

4.46 Parameter EthEthTrcvDriverVendorIdAndApiInfix

The vendor id and vendor api infix for the EthTrcv driver, if any.

This is used to include the correct header file and to point to the EthTrcv callbacks.

Example: 43_PHYGROUPB for a 43_PHYGROUPB driver or leave it blank to include EthTrcv.h

Property	Value
type	ECUC-STRING-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-LINK-TIME: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE
defaultValue	

4.47 Parameter EthEthSwtDriverVendorIdAndApiInfix

The vendor id and vendor api infix for the EthSwt driver, if any.

This is used to include the correct header file and to point to the EthSwt APIs.

Example: 43_SJA1110 for a 43_SJA1110 driver or leave it blank to include EthSwt.h

Property	Value
type	ECUC-STRING-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-LINK-TIME: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE
defaultValue	

4.48 Parameter EthDuplexMode

Defines the controller's duplex mode.

Property	Value
type	ECUC-ENUMERATION-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-LINK-TIME: LINK
	VARIANT-POST-BUILD: POST-BUILD
defaultValue	ETH_FULL_DUPLEX
literals	['ETH_FULL_DUPLEX', 'ETH_HALF_DUPLEX']

4.49 Parameter EthTimeStampRequiredAccuracy

Value given in nanoseconds.

Represents the minimum required accuracy for this gPTP device.

To ensure there's enough headroom for clock drifts from both the GM and Slave, a maximum drift percentage per host must be assumed.

That is, TotalDriftPercentage = MaximumGmDriftPercentage + MaximumSlaveDriftPercentage. Then the following inequality must be be true:

EthTimeStampReferenceClock * (1 + TotalDriftPercentage / 100) > (1 / EthTimeStampRequiredAccuracy) * 10^9.

Property	Value
type	ECUC-INTEGER-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	0
upperMultiplicity	1
postBuildVariantMultiplicity	true
multiplicityConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-LINK-TIME: LINK
	VARIANT-POST-BUILD: POST-BUILD
postBuildVariantValue	true

Property	Value
valueConfigClasses	VARIANT-LINK-TIME: LINK
	VARIANT-POST-BUILD: POST-BUILD
	VARIANT-PRE-COMPILE: PRE-COMPILE
defaultValue	40
max	4294967295
min	1

4.50 Reference EthModuleReferenceClock

Reference to the module clock for the GMAC.

Property	Value
type	ECUC-REFERENCE-DEF
origin	NXP
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-LINK-TIME: LINK
	VARIANT-POST-BUILD: POST-BUILD
	VARIANT-PRE-COMPILE: PRE-COMPILE
requiresSymbolicNameValue	False
destination	/AUTOSAR/EcucDefs/Mcu/McuModuleConfiguration/McuClockSetting↩ Config/McuClockReferencePoint

4.51 Reference EthTimeStampReferenceClock

Reference to the Timestamp source clock for the GMAC.

Property	Value
type	ECUC-REFERENCE-DEF
origin	NXP
lowerMultiplicity	0
upperMultiplicity	1
postBuildVariantMultiplicity	true
multiplicityConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-LINK-TIME: LINK
	VARIANT-POST-BUILD: POST-BUILD
postBuildVariantValue	true

Property	Value
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-LINK-TIME: LINK
	VARIANT-POST-BUILD: POST-BUILD
requiresSymbolicNameValue	False
destination	/AUTOSAR/EcucDefs/Mcu/McuModuleConfiguration/McuClockSetting↩ Config/McuClockReferencePoint

4.52 Container EthCtrlConfigSafety

Safety Controller Configuration.

Included subcontainers:

- None

Property	Value
type	ECUC-PARAM-CONF-CONTAINER-DEF
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A

4.53 Parameter EthCtrlSafetyCallback

Callback function invoked when a safety event is encountered.

Property	Value
type	ECUC-FUNCTION-NAME-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-LINK-TIME: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE
defaultValue	Eth_Ipw_SafetyIrqCallback

4.54 Parameter EthCtrlEnableEcc

Enables/Disables SECDED data and address protection for MTL and EST memories.

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-LINK-TIME: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE
defaultValue	false

4.55 Parameter EthCtrlEnableDpp

Enables/Disables parity protection for Rx and Tx data paths.

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-LINK-TIME: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE
defaultValue	false

4.56 Parameter EthCtrlEnableFsmSpp

Enables/Disables state parity protection for the internal FSM.

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-LINK-TIME: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE
defaultValue	false

4.57 Parameter EthCtrlEnableFsmTimeout

Enables/Disables timeout protection for the internal FSM.

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-LINK-TIME: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE
defaultValue	false

4.58 Reference ETH_E_SAFETY

Reference to the DemEventParameter which shall be issued when any safety error has occurred.

Property	Value
type	ECUC-REFERENCE-DEF
origin	NXP
lowerMultiplicity	1

Property	Value
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-LINK-TIME: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE
requiresSymbolicNameValue	true
destination	/AUTOSAR/EcucDefs/Dem/DemConfigSet/DemEventParameter

4.59 Container EthCtrlConfigMac

MAC Controller Configuration.

Included subcontainers:

- None

Property	Value
type	ECUC-PARAM-CONF-CONTAINER-DEF
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A

4.60 Parameter MAC_CONFIG_CRC_STRIPPING

The last four bytes (FCS) of all packets of Ether type (type field greater than 1,536) are stripped and dropped before forwarding the packet to the application.

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A

Property	Value
postBuildVariantValue	false
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-LINK-TIME: LINK
	VARIANT-POST-BUILD: POST-BUILD
defaultValue	true

4.61 Parameter MAC_CONFIG_AUTO_PAD

The MAC strips the Pad or FCS field on the incoming packets only if the value of the length field is less than 1,536 bytes.

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-LINK-TIME: LINK
	VARIANT-POST-BUILD: POST-BUILD
defaultValue	true

4.62 Parameter MAC_CONFIG_JUMBO_PKT_EN

The MAC allows jumbo packets of 9,018 bytes (9,022 bytes for VLAN tagged packets).

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-LINK-TIME: LINK

Property	Value
	VARIANT-POST-BUILD: POST-BUILD
defaultValue	false

4.63 Parameter MAC_CONFIG_LOOPBACK

MAC operates in the loopback mode at GMII or MII. The Rx clock input (clk_rx_i) is required for the loopback to work properly.

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-LINK-TIME: LINK
	VARIANT-POST-BUILD: POST-BUILD
defaultValue	false

4.64 Parameter MAC_CONFIG_ENABLE_CRS_FD

The MAC transmitter checks the CRS signal before packet transmission in the full-duplex mode. The MAC starts the transmission only when the CRS signal is low.

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-LINK-TIME: LINK
	VARIANT-POST-BUILD: POST-BUILD
defaultValue	false

4.65 Parameter MAC_CONFIG_DISABLE_RECEIVE_OWN_HD

The MAC disables the reception of packets when the gmii_txen_o is asserted in the half-duplex mode.

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-LINK-TIME: LINK
	VARIANT-POST-BUILD: POST-BUILD
defaultValue	false

4.66 Parameter MAC_CONFIG_DISABLE_CRS_HD

The MAC transmitter ignores the (G)MII CRS signal during packet transmission in the half-duplex mode. As a result, no errors are generated because of Loss of Carrier or No Carrier during transmission.

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-LINK-TIME: LINK
	VARIANT-POST-BUILD: POST-BUILD
defaultValue	false

4.67 Parameter MAC_CONFIG_DISABLE_RETRY

The MAC attempts only one transmission. When a collision occurs on the (G)MII interface, the MAC ignores the current packet transmission and reports a Packet Abort with excessive collision error in the Tx packet status.

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-LINK-TIME: LINK
	VARIANT-POST-BUILD: POST-BUILD
defaultValue	false

4.68 Parameter MAC_CONFIG_DEFERRAL_CHECK_HD

The deferral check function is enabled in the MAC. The MAC issues a Packet Abort status, along with the excessive deferral error bit set in the Tx packet status, when the Tx state machine is deferred for more than 24,288 bit times in 10/100 Mbps mode (155,680 bits times in 1000/2500 Mbps operation). Deferral begins when the transmitter is ready to transmit, but it is prevented because of an active carrier sense signal (CRS) on (G)MII.

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-LINK-TIME: LINK
	VARIANT-POST-BUILD: POST-BUILD
defaultValue	false

4.69 Container EthCtrlConfigPacketFilter

Packet Filtering Configuration.

Included subcontainers:

- None

Property	Value
type	ECUC-PARAM-CONF-CONTAINER-DEF
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A

4.70 Parameter PKT_FILTER_RECV_ALL

Passes all received packets to the application, irrespective of whether they pass the address filter or not.

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-LINK-TIME: LINK
	VARIANT-POST-BUILD: POST-BUILD
defaultValue	false

4.71 Parameter PKT_FILTER_HASH_OR_PERFECT_FILTER

The address filter passes a packet if it matches either the perfect filtering or hash filtering.

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF

Property	Value
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-LINK-TIME: LINK
	VARIANT-POST-BUILD: POST-BUILD
defaultValue	true

4.72 Parameter PKT_FILTER_BLOCK_CONTROL_PKTS

Blocks all control packets from reaching the application.

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-LINK-TIME: LINK
	VARIANT-POST-BUILD: POST-BUILD
defaultValue	false

4.73 Parameter PKT_FILTER_BLOCK_PAUSE_PKTS

Forwards all control packets except Pause packets to the application even if they fail the Address filter.

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1

Property	Value
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-LINK-TIME: LINK
	VARIANT-POST-BUILD: POST-BUILD
defaultValue	false

4.74 Parameter PKT_FILTER_PASS_ALL_CONTROL_PKTS

Forwards all control packets to the application even if they fail the Address filter.

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-LINK-TIME: LINK
	VARIANT-POST-BUILD: POST-BUILD
defaultValue	false

4.75 Parameter PKT_FILTER_PASS_CONTROL_PKTS_ADDR_MATCH

Forwards the control packets that pass the Address filter.

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A

Property	Value
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-LINK-TIME: LINK
	VARIANT-POST-BUILD: POST-BUILD
defaultValue	false

4.76 Parameter PKT_FILTER_DISABLE_BROADCAST

Blocks all incoming broadcast packets.

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-LINK-TIME: LINK
	VARIANT-POST-BUILD: POST-BUILD
defaultValue	false

4.77 Parameter PKT_FILTER_PASS_ALL_MULTICAST

All received packets with a multicast destination address are passed.

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
	VARIANT-PRE-COMPILE: PRE-COMPILE

valueConfigClasses

Property	Value
	VARIANT-LINK-TIME: LINK
	VARIANT-POST-BUILD: POST-BUILD
defaultValue	false

4.78 Parameter PKT_FILTER_DST_ADDR_INV_FILTER_EN

The Address checking operates in inverse filtering mode for the destination address comparison for both unicast and multicast packets.

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-LINK-TIME: LINK
	VARIANT-POST-BUILD: POST-BUILD
defaultValue	false

4.79 Parameter PKT_FILTER_HASH_MULTICAST

The destination address filtering of received multicast packets is done according to the hash table.

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-LINK-TIME: LINK
	VARIANT-POST-BUILD: POST-BUILD
defaultValue	false

4.80 Parameter PKT_FILTER_HASH_UNICAST

The destination address filtering of received unicast packets is done according to the hash table.

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-LINK-TIME: LINK
	VARIANT-POST-BUILD: POST-BUILD
defaultValue	false

4.81 Parameter PKT_FILTER_PROMISCUOUS_MODE

Passes all incoming packets irrespective of the destination or source address.

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-LINK-TIME: LINK
	VARIANT-POST-BUILD: POST-BUILD
defaultValue	false

4.82 Container EthCtrlConfigEgress

Controller Configuration Egress: Configuration of one Ethernet controller egress behavior.

Included subcontainers:

- EthCtrlConfigEgressFifo
- EthCtrlConfigScheduler
- EthCtrlConfigShaper

Property	Value
type	ECUC-PARAM-CONF-CONTAINER-DEF
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A

4.83 Reference EthCtrlConfigEgressLastSchedulerRef

Reference to the scheduler which is the last in the Egress structure.

Property	Value
type	ECUC-REFERENCE-DEF
origin	AUTOSAR_ECUC
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-PRE-COMPILE: PRE-COMPILE
	VARIANT-LINK-TIME: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
requiresSymbolicNameValue	False
destination	/AUTOSAR/EcucDefs/Eth/EthConfigSet/EthCtrlConfig/EthCtrlConfig↩ Egress/EthCtrlConfigScheduler

4.84 Container EthCtrlConfigEgressFifo

Controller Configuration Egress: Represents a FIFO at the egress side.

Included subcontainers:

- None

Property	Value
type	ECUC-PARAM-CONF-CONTAINER-DEF
lowerMultiplicity	0
upperMultiplicity	2
postBuildVariantMultiplicity	true
multiplicityConfigClasses	VARIANT-LINK-TIME: LINK
	VARIANT-POST-BUILD: POST-BUILD
	VARIANT-PRE-COMPILE: PRE-COMPILE

4.85 Parameter EthCtrlConfigEgressFifoBufLenByte

Length of FIFO elements in bytes.

Note that this represents the length of a single packet buffer (including the 14-bytes Ethernet frame header and the 4-bytes FCS).

Therefore, the payload length will be equal to EthCtrlConfigEgressFifoBufLenByte - 18.

Property	Value
type	ECUC-INTEGER-PARAM-DEF
origin	AUTOSAR_ECUC
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-LINK-TIME: LINK
	VARIANT-POST-BUILD: POST-BUILD
	VARIANT-PRE-COMPILE: PRE-COMPILE
defaultValue	64
max	1536
min	64

4.86 Parameter EthCtrlConfigEgressFifoBufTotal

FIFO Buffer Count (defines the number of packet buffers of length EthCtrlConfigEgressFifoBufLenByte).

Property	Value
type	ECUC-INTEGER-PARAM-DEF
origin	AUTOSAR_ECUC

Property	Value
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-LINK-TIME: LINK
	VARIANT-POST-BUILD: POST-BUILD
	VARIANT-PRE-COMPILE: PRE-COMPILE
defaultValue	16
max	128
min	4

4.87 Parameter EthCtrlConfigEgressFifoIdx

Egress FIFO index.

Property	Value
type	ECUC-INTEGER-PARAM-DEF
origin	AUTOSAR_ECUC
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-LINK-TIME: LINK
	VARIANT-POST-BUILD: POST-BUILD
	VARIANT-PRE-COMPILE: PRE-COMPILE
defaultValue	0
max	1
min	0

4.88 Parameter EthCtrlConfigEgressFifoCallback

Callback function invoked when an Egress (Tx) FIFO event is encountered.

Property	Value
type	ECUC-FUNCTION-NAME-DEF
origin	NXP

Property	Value
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-LINK-TIME: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE
defaultValue	Eth_TxIrqCallback

4.89 Parameter EthCtrlConfigEgressFifoPriorityAssignment

Message egress prority assignment.

Property	Value
type	ECUC-INTEGER-PARAM-DEF
origin	AUTOSAR_ECUC
symbolicNameValue	false
lowerMultiplicity	0
upperMultiplicity	7
postBuildVariantMultiplicity	true
multiplicityConfigClasses	VARIANT-LINK-TIME: LINK
	VARIANT-POST-BUILD: POST-BUILD
	VARIANT-PRE-COMPILE: PRE-COMPILE
postBuildVariantValue	true
valueConfigClasses	VARIANT-LINK-TIME: LINK
	VARIANT-POST-BUILD: POST-BUILD
	VARIANT-PRE-COMPILE: PRE-COMPILE
defaultValue	0
max	7
min	0

4.90 Container EthCtrlConfigScheduler

Controller Configuration Scheduler: Represents a Scheduler on the Egress side.

Included subcontainers:

- EthCtrlConfigSchedulerPredecessor
- EthCtrlConfigSchedulerVendorSpecific

Property	Value
type	ECUC-PARAM-CONF-CONTAINER-DEF
lowerMultiplicity	1
upperMultiplicity	255
postBuildVariantMultiplicity	true
multiplicityConfigClasses	VARIANT-LINK-TIME: LINK
	VARIANT-POST-BUILD: POST-BUILD
	VARIANT-PRE-COMPILE: PRE-COMPILE

4.91 Container EthCtrlConfigSchedulerPredecessor

Controller Configuration Scheduler Predecessors: Defines an ordered list of predecessors for this scheduler.

Included subcontainers:

- EthCtrlConfigSchedulerPredecessorVendorSpecific

Property	Value
type	ECUC-PARAM-CONF-CONTAINER-DEF
lowerMultiplicity	1
upperMultiplicity	Infinite
postBuildVariantMultiplicity	true
multiplicityConfigClasses	VARIANT-LINK-TIME: LINK
	VARIANT-POST-BUILD: POST-BUILD
	VARIANT-PRE-COMPILE: PRE-COMPILE

4.92 Parameter EthCtrlConfigSchedulerPredecessorOrder

Defines the order of the scheduler predecessors.

Property	Value
type	ECUC-INTEGER-PARAM-DEF
origin	AUTOSAR_ECUC
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-LINK-TIME: LINK

Property	Value
	VARIANT-POST-BUILD: POST-BUILD
	VARIANT-PRE-COMPILE: PRE-COMPILE
defaultValue	0
max	255
min	0

4.93 Reference EthCtrlConfigSchedulerPredecessorRef

Choice reference to the scheduler predecessor.

Property	Value
type	ECUC-CHOICE-REFERENCE-DEF
origin	AUTOSAR_ECUC
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-LINK-TIME: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE
requiresSymbolicNameValue	False
destinations	['/AUTOSAR/EcucDefs/Eth/EthConfigSet/EthCtrlConfig/EthCtrlConfig↩ Egress/EthCtrlConfigEgressFifo', '/AUTOSAR/EcucDefs/Eth/EthConfigSet/↩ EthCtrlConfig/EthCtrlConfigEgress/EthCtrlConfigScheduler', '/AUTOSA↩ R/EcucDefs/Eth/EthConfigSet/EthCtrlConfig/EthCtrlConfigEgress/EthCtrl↩ ConfigShaper']

4.94 Container EthCtrlConfigSchedulerPredecessorVendorSpecific

Scheduler Predecessor vendor specific configuration.

Included subcontainers:

- None

Property	Value
type	ECUC-PARAM-CONF-CONTAINER-DEF
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A

4.95 Parameter EthQueueBandwidth

Represents the bandwidth percentage allocated for this queue.

In SP, this parameter is irrelevant.

In WRR, EthQueueBandwidth / 100 * EthTotalPacketsPerCycle must not exceed 100 packets.

In DWRR, EthQueueBandwidth / 100 * EthTotalQuantumPerCycle must not exceed 1,250,000 bytes.

This field is editable only when the scheduling algorithm is set to anything other than STRICT_PRIORITY.

Property	Value
type	ECUC-INTEGER-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-LINK-TIME: LINK
	VARIANT-POST-BUILD: POST-BUILD
	VARIANT-PRE-COMPILE: PRE-COMPILE
defaultValue	100
max	100
min	1

4.96 Container EthCtrlConfigSchedulerVendorSpecific

Scheduler vendor specific configuration.

Included subcontainers:

- None

Property	Value
type	ECUC-PARAM-CONF-CONTAINER-DEF
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A

4.97 Parameter EthTxSchedulerAlgorithm

Configures the global Tx Scheduling Algorithm affecting all enabled Tx Queues.

For both round-robin algorithms, the queues are serviced in Qn > Qn-1 > ... > Q0 order based on their allocated bandwidths.

When Credit-Based Shapers are configured in EthCtrlConfigSchedulerPredecessor, the scheduling algorithm must be set to STRICT_PRIORITY.

In a single Egress FIFO configuration, the scheduling algorithm must be set to STRICT_PRIORITY.

Property	Value
type	ECUC-ENUMERATION-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-LINK-TIME: LINK
	VARIANT-POST-BUILD: POST-BUILD
	VARIANT-PRE-COMPILE: PRE-COMPILE
defaultValue	STRICT_PRIORITY
literals	['STRICT_PRIORITY', 'WEIGHTED_ROUND_ROBIN', 'WEIGHTED_F↩ AIR_QUEUING', 'DEFICIT_WEIGHTED_ROUND_ROBIN']

4.98 Parameter EthTotalPacketsPerCycle

Configures the total weight value (in packets) per arbitration cycle from which individual queue weights are calculated based on their allocated bandwidths.

For example, if queues (Q2, Q1, Q0) are configured with bandwidths (50%, 25%, 25%) and EthTotalPacketsPerCycle = 4 packets,

then the individual queue weights are (4 * 0.5, 4 * 0.25, 4 * 0.25) = (2, 1, 1) packets.

This field is editable only when the scheduling algorithm is set to WEIGHTED_ROUND_ROBIN.

Property	Value
type	ECUC-INTEGER-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1

Property	Value
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-LINK-TIME: LINK
	VARIANT-POST-BUILD: POST-BUILD
	VARIANT-PRE-COMPILE: PRE-COMPILE
defaultValue	10
max	50000000
min	1

4.99 Parameter EthTotalQuantumPerCycle

Configures the total quantum value (in bytes) per arbitration cycle from which individual queue quantums are calculated based on their allocated bandwidths.

For example, if queues (Q2, Q1, Q0) are configured with bandwidths (50%, 25%, 25%) and EthTotalQuantumPerCycle = 2000 bytes,

then the individual queue quantums are (2000 * 0.5, 2000 * 0.25, 2000 * 0.25) = (1000, 500, 500) bytes.

This field is editable only when the scheduling algorithm is set to DEFICIT_WEIGHTED_ROUND_ROBIN.

Property	Value
type	ECUC-INTEGER-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-LINK-TIME: LINK
	VARIANT-POST-BUILD: POST-BUILD
	VARIANT-PRE-COMPILE: PRE-COMPILE
defaultValue	1000
max	4294967295
min	1

4.100 Container EthCtrlConfigShaper

Controller Configuration Shaper : Represents a Shaper at the Egress side.

Included subcontainers:

- EthCtrlConfigShaperVendorSpecific

Property	Value
type	ECUC-PARAM-CONF-CONTAINER-DEF
lowerMultiplicity	0
upperMultiplicity	2
postBuildVariantMultiplicity	true
multiplicityConfigClasses	VARIANT-LINK-TIME: LINK
	VARIANT-POST-BUILD: POST-BUILD
	VARIANT-PRE-COMPILE: PRE-COMPILE

4.101 Parameter EthCtrlConfigShaperIdleSlope

Defines the increase of credit in bits per second for the AVB shaper.

Property	Value
type	ECUC-INTEGER-PARAM-DEF
origin	AUTOSAR_ECUC
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-LINK-TIME: LINK
	VARIANT-POST-BUILD: POST-BUILD
	VARIANT-PRE-COMPILE: PRE-COMPILE
defaultValue	0
max	150000000
min	0

4.102 Reference EthCtrlConfigShaperPredecessorFifoRef

Reference to the FIFO which is the predecessor for this Shaper.

Property	Value
type	ECUC-REFERENCE-DEF
origin	AUTOSAR_ECUC

Property	Value
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-LINK-TIME: LINK
	VARIANT-POST-BUILD: POST-BUILD
	VARIANT-PRE-COMPILE: PRE-COMPILE
requiresSymbolicNameValue	False
destination	/AUTOSAR/EcucDefs/Eth/EthConfigSet/EthCtrlConfig/EthCtrlConfig↩ Egress/EthCtrlConfigEgressFifo

4.103 Container EthCtrlConfigShaperVendorSpecific

Shaper vendor specific configuration.

Included subcontainers:

- None

Property	Value
type	ECUC-PARAM-CONF-CONTAINER-DEF
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A

4.104 Parameter EthCtrlConfigHiCredit

Defines the maximum value (in bits) that can be accumulated in the credit parameter.

This value should be equal to the maximum size (in bits) of an interfering (best-effort) frame.

Property	Value
type	ECUC-INTEGER-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1

Property	Value
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-LINK-TIME: LINK
	VARIANT-POST-BUILD: POST-BUILD
	VARIANT-PRE-COMPILE: PRE-COMPILE
defaultValue	12000
max	131072
min	0

4.105 Parameter EthCtrlConfigLoCredit

Defines the minimum value (in bits) that can be accumulated in the credit parameter.

This value should be equal to twice the maximum size (in bits) of a frame transmitted from this queue.

Property	Value
type	ECUC-INTEGER-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-LINK-TIME: LINK
	VARIANT-POST-BUILD: POST-BUILD
	VARIANT-PRE-COMPILE: PRE-COMPILE
defaultValue	4096
max	131072
min	0

4.106 Container EthCtrlConfigTimeAwareShaper

Controller Time Aware Shaper: Represents Time Aware Shaper configuration on the Egress side.

Included subcontainers:

- EthCtrlConfigTimeAwareShaper

Property	Value
type	ECUC-PARAM-CONF-CONTAINER-DEF
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A

4.107 Container EthCtrlConfigTimeAwareShaper

Controller Time Aware Shaper: Represents Time Aware Shaper configuration on the Egress side.

Included subcontainers:

- EthGateControlList
- EthPreemptionClassification

Property	Value
type	ECUC-PARAM-CONF-CONTAINER-DEF
lowerMultiplicity	0
upperMultiplicity	0
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A

4.108 Parameter EthGateConfigBaseTimeInSecond

The start time for the Gate control list

This value in second.

Once the execution of a given list begins, the implementation can update this value to indicate the next list execution begin time.

Property	Value
type	ECUC-INTEGER-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A

Property	Value
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-LINK-TIME: LINK
	VARIANT-POST-BUILD: POST-BUILD
	VARIANT-PRE-COMPILE: PRE-COMPILE
defaultValue	1000
max	4294967295
min	0

4.109 Parameter EthGateConfigBaseTimeInNanoSecond

The start time for the Gate control list

This value in nanosecond.

Once the execution of a given list begins, the implementation can update this value to indicate the next list execution begin time.

Property	Value
type	ECUC-INTEGER-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-LINK-TIME: LINK
	VARIANT-POST-BUILD: POST-BUILD
	VARIANT-PRE-COMPILE: PRE-COMPILE
defaultValue	1000
max	4294967295
min	0

4.110 Parameter EthGateConfigCycleTime

The time at which the execution of the gate control list should be repeated.

This value in nanosecond.

Property	Value
type	ECUC-INTEGER-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-LINK-TIME: LINK
	VARIANT-POST-BUILD: POST-BUILD
	VARIANT-PRE-COMPILE: PRE-COMPILE
defaultValue	0
max	259294967295
min	0

4.111 Parameter EthGateConfigExtendTime

The parameter holds the amount of time (in nanoseconds) can be extended before switching to the new Gate Control List.

This will be useful in avoiding small fragments of the current list being executed before switching to a new list.

Property	Value
type	ECUC-INTEGER-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-LINK-TIME: LINK
	VARIANT-POST-BUILD: POST-BUILD
	VARIANT-PRE-COMPILE: PRE-COMPILE
defaultValue	0
max	2147483647
min	0

4.112 Parameter EthCtrlEnablePreemtion

Enabled - Frame preemption is enabled.

Disabled - Frame preemption is disabled.

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-LINK-TIME: LINK
	VARIANT-POST-BUILD: POST-BUILD
	VARIANT-PRE-COMPILE: PRE-COMPILE
defaultValue	false

4.113 Parameter EthReleaseAdvanceTime

The maximum time in nanoseconds elapse between issuing a RELEASE to the MAC and the MAC being ready to resume

transmission of preemptable frames.

Property	Value
type	ECUC-INTEGER-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-LINK-TIME: LINK
	VARIANT-POST-BUILD: POST-BUILD
	VARIANT-PRE-COMPILE: PRE-COMPILE
defaultValue	1000
max	65535
min	1

4.114 Parameter EthHoldAdvanceTime

The maximum time in nanoseconds elapse between issuing a HOLD to the MAC and the MAC ceasing to transmit

any preemptable frame.

Property	Value
type	ECUC-INTEGER-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-LINK-TIME: LINK
	VARIANT-POST-BUILD: POST-BUILD
	VARIANT-PRE-COMPILE: PRE-COMPILE
defaultValue	1000
max	65535
min	1

4.115 Container EthGateControlList

Gate Control List : governs the gate close or open events based on schedule provided.

Included subcontainers:

- EthGateControlFifo

Property	Value
type	ECUC-PARAM-CONF-CONTAINER-DEF
lowerMultiplicity	0
upperMultiplicity	256
postBuildVariantMultiplicity	true
multiplicityConfigClasses	VARIANT-LINK-TIME: LINK
	VARIANT-POST-BUILD: POST-BUILD
	VARIANT-PRE-COMPILE: PRE-COMPILE

4.116 Parameter EthCtrlConfigInterval

The time in nano seconds for gate controls are valid and should be applied before reading the next gate controls from the list.

Property	Value
type	ECUC-INTEGER-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-LINK-TIME: LINK
	VARIANT-POST-BUILD: POST-BUILD
	VARIANT-PRE-COMPILE: PRE-COMPILE
defaultValue	0
max	16000000
min	0

4.117 Container EthGateControlFifo

Gate control fifo: Configurate Fifo is open or close in duration interval.

Included subcontainers:

- None

Property	Value
type	ECUC-PARAM-CONF-CONTAINER-DEF
lowerMultiplicity	0
upperMultiplicity	2
postBuildVariantMultiplicity	true
multiplicityConfigClasses	VARIANT-LINK-TIME: LINK
	VARIANT-POST-BUILD: POST-BUILD
	VARIANT-PRE-COMPILE: PRE-COMPILE

4.118 Parameter EthCtrlConfigGateStatus

This parameters to select gate status is open or close.

Property	Value
type	ECUC-ENUMERATION-PARAM-DEF
origin	NXP

Property	Value
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-LINK-TIME: LINK
	VARIANT-POST-BUILD: POST-BUILD
	VARIANT-PRE-COMPILE: PRE-COMPILE
defaultValue	OPEN
literals	['CLOSE', 'OPEN']

4.119 Reference EthCtrlConfigFifoRef

Reference to the FIFO which is the predecessor for this time aware shaper.

Property	Value
type	ECUC-REFERENCE-DEF
origin	NXP
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-LINK-TIME: LINK
	VARIANT-POST-BUILD: POST-BUILD
	VARIANT-PRE-COMPILE: PRE-COMPILE
requiresSymbolicNameValue	False
destination	/AUTOSAR/EcucDefs/Eth/EthConfigSet/EthCtrlConfig/EthCtrlConfig↩ Egress/EthCtrlConfigEgressFifo

4.120 Container EthPreemptionClassification

Preemption claissification : select reference fifo is preemptable or express.

Included subcontainers:

- None

Property	Value
type	ECUC-PARAM-CONF-CONTAINER-DEF
lowerMultiplicity	0
upperMultiplicity	2
postBuildVariantMultiplicity	true
multiplicityConfigClasses	VARIANT-LINK-TIME: LINK
	VARIANT-POST-BUILD: POST-BUILD
	VARIANT-PRE-COMPILE: PRE-COMPILE

4.121 Parameter EthCtrlConfigPreemptionClassification

This parameters to select the queue is preemptable or express

Property	Value
type	ECUC-ENUMERATION-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-LINK-TIME: LINK
	VARIANT-POST-BUILD: POST-BUILD
	VARIANT-PRE-COMPILE: PRE-COMPILE
defaultValue	EXPRESS
literals	['PREEMPTABLE', 'EXPRESS']

4.122 Reference EthPreemptionFifoRef

Reference to the FIFO which is the predecessor for preemption.

Property	Value
type	ECUC-REFERENCE-DEF
origin	NXP
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false

Property	Value
valueConfigClasses	VARIANT-LINK-TIME: LINK
	VARIANT-POST-BUILD: POST-BUILD
	VARIANT-PRE-COMPILE: PRE-COMPILE
requiresSymbolicNameValue	False
destination	/AUTOSAR/EcucDefs/Eth/EthConfigSet/EthCtrlConfig/EthCtrlConfig↩ Egress/EthCtrlConfigEgressFifo

4.123 Container EthCtrlConfigIngress

Controller Configuration Ingress: Configuration of one Ethernet controller ingress behavior.

Included subcontainers:

- EthCtrlConfigIngressFifo

Property	Value
type	ECUC-PARAM-CONF-CONTAINER-DEF
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A

4.124 Container EthCtrlConfigIngressFifo

Controller Configuration Ingress : Represents a FIFO at the ingress side.

Included subcontainers:

- None

Property	Value
type	ECUC-PARAM-CONF-CONTAINER-DEF
lowerMultiplicity	0
upperMultiplicity	2
postBuildVariantMultiplicity	true
multiplicityConfigClasses	VARIANT-LINK-TIME: LINK
	VARIANT-POST-BUILD: POST-BUILD
	VARIANT-PRE-COMPILE: PRE-COMPILE

4.125 Parameter EthCtrlConfigIngressFifoBufLenByte

Length of FIFO elements in bytes.

Note that this represents the length of a single packet buffer (including the 14-bytes Ethernet frame header and the 4-bytes FCS).

Therefore, the payload length will be equal to EthCtrlConfigIngressFifoBufLenByte - 18.

Property	Value
type	ECUC-INTEGER-PARAM-DEF
origin	AUTOSAR_ECUC
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-LINK-TIME: LINK
	VARIANT-POST-BUILD: POST-BUILD
	VARIANT-PRE-COMPILE: PRE-COMPILE
defaultValue	64
max	1536
min	64

4.126 Parameter EthCtrlConfigIngressFifoBufTotal

FIFO Buffer Count (defines the number of packet buffers of length EthCtrlConfigIngressFifoBufLenByte).

Property	Value
type	ECUC-INTEGER-PARAM-DEF
origin	AUTOSAR_ECUC
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-LINK-TIME: LINK
	VARIANT-POST-BUILD: POST-BUILD
	VARIANT-PRE-COMPILE: PRE-COMPILE
defaultValue	16
max	128
min	4

4.127 Parameter EthCtrlConfigIngressFifoIdx

Ingress FIFO index.

Property	Value
type	ECUC-INTEGER-PARAM-DEF
origin	AUTOSAR_ECUC
symbolicNameValue	true
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-LINK-TIME: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE
defaultValue	0
max	1
min	0

4.128 Parameter EthCtrlConfigIngressFifoCallback

Callback function invoked when an Ingress (Rx) FIFO event is encountered.

Property	Value
type	ECUC-FUNCTION-NAME-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-LINK-TIME: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE
defaultValue	Eth_RxIrqCallback

4.129 Parameter EthCtrlConfigIngressFifoPriorityAssignment

Message ingress prority assignment.

Property	Value
type	ECUC-INTEGER-PARAM-DEF
origin	AUTOSAR_ECUC
symbolicNameValue	false
lowerMultiplicity	0
upperMultiplicity	7
postBuildVariantMultiplicity	true
multiplicityConfigClasses	VARIANT-LINK-TIME: LINK
	VARIANT-POST-BUILD: POST-BUILD
	VARIANT-PRE-COMPILE: PRE-COMPILE
postBuildVariantValue	true
valueConfigClasses	VARIANT-LINK-TIME: LINK
	VARIANT-POST-BUILD: POST-BUILD
	VARIANT-PRE-COMPILE: PRE-COMPILE
defaultValue	0
max	7
min	0

4.130 Container EthDemEventParameterRefs

Container for the references to DemEventParameter elements which shall be invoked using the API Dem_ReportErrorStatus in case the corresponding error occurs.

The EventId is taken from the referenced DemEventParameter's DemEventId value.

The standardized errors are provided in the container and can be extended by vendor specific error references.

Container is editable only when EthGeneral/EthGeneralVendorSpecific/EthDisableDemEventDetect = FALSE.

Included subcontainers:

- None

Property	Value
type	ECUC-PARAM-CONF-CONTAINER-DEF
lowerMultiplicity	0
upperMultiplicity	1
postBuildVariantMultiplicity	true
multiplicityConfigClasses	VARIANT-LINK-TIME: LINK
	VARIANT-POST-BUILD: POST-BUILD
	VARIANT-PRE-COMPILE: PRE-COMPILE

4.131 Reference ETH_E_ACCESS

Reference to the DemEventParameter which shall be issued when the error 'Controller access failed' has occurred.

Property	Value
type	ECUC-REFERENCE-DEF
origin	AUTOSAR_ECUC
lowerMultiplicity	0
upperMultiplicity	1
postBuildVariantMultiplicity	true
multiplicityConfigClasses	VARIANT-LINK-TIME: LINK
	VARIANT-POST-BUILD: POST-BUILD
	VARIANT-PRE-COMPILE: PRE-COMPILE
postBuildVariantValue	true
valueConfigClasses	VARIANT-LINK-TIME: LINK
	VARIANT-POST-BUILD: POST-BUILD
	VARIANT-PRE-COMPILE: PRE-COMPILE
requiresSymbolicNameValue	true
destination	/AUTOSAR/EcucDefs/Dem/DemConfigSet/DemEventParameter

4.132 Reference ETH_E_RX_FRAMES_LOST

Reference to the DemEventParameter which shall be issued when the error 'Rx Frame Lost' has occurred.

Property	Value
type	ECUC-REFERENCE-DEF
origin	AUTOSAR_ECUC
lowerMultiplicity	0
upperMultiplicity	1
postBuildVariantMultiplicity	true
multiplicityConfigClasses	VARIANT-LINK-TIME: LINK
	VARIANT-POST-BUILD: POST-BUILD
	VARIANT-PRE-COMPILE: PRE-COMPILE
postBuildVariantValue	true
valueConfigClasses	VARIANT-LINK-TIME: LINK
	VARIANT-POST-BUILD: POST-BUILD
	VARIANT-PRE-COMPILE: PRE-COMPILE
requiresSymbolicNameValue	true
destination	/AUTOSAR/EcucDefs/Dem/DemConfigSet/DemEventParameter

4.133 Reference ETH_E_CRC

Reference to the DemEventParameter which shall be issued when the error 'CRC Failure' has occurred.

Property	Value
type	ECUC-REFERENCE-DEF
origin	AUTOSAR_ECUC
lowerMultiplicity	0
upperMultiplicity	1
postBuildVariantMultiplicity	true
multiplicityConfigClasses	VARIANT-LINK-TIME: LINK
	VARIANT-POST-BUILD: POST-BUILD
	VARIANT-PRE-COMPILE: PRE-COMPILE
postBuildVariantValue	true
valueConfigClasses	VARIANT-LINK-TIME: LINK
	VARIANT-POST-BUILD: POST-BUILD
	VARIANT-PRE-COMPILE: PRE-COMPILE
requiresSymbolicNameValue	true
destination	/AUTOSAR/EcucDefs/Dem/DemConfigSet/DemEventParameter

4.134 Reference ETH_E_UNDERSIZEFRAME

Reference to the DemEventParameter which shall be issued when the error 'Undersized Frame' has occurred.

Property	Value
type	ECUC-REFERENCE-DEF
origin	AUTOSAR_ECUC
lowerMultiplicity	0
upperMultiplicity	1
postBuildVariantMultiplicity	true
multiplicityConfigClasses	VARIANT-LINK-TIME: LINK
	VARIANT-POST-BUILD: POST-BUILD
	VARIANT-PRE-COMPILE: PRE-COMPILE
postBuildVariantValue	true
valueConfigClasses	VARIANT-LINK-TIME: LINK
	VARIANT-POST-BUILD: POST-BUILD
	VARIANT-PRE-COMPILE: PRE-COMPILE
requiresSymbolicNameValue	true
destination	/AUTOSAR/EcucDefs/Dem/DemConfigSet/DemEventParameter

4.135 Reference ETH_E_OVERSIZEFRAME

Reference to the DemEventParameter which shall be issued when the error 'Oversized Frame' has occurred.

Property	Value
type	ECUC-REFERENCE-DEF
origin	AUTOSAR_ECUC
lowerMultiplicity	0
upperMultiplicity	1
postBuildVariantMultiplicity	true
multiplicityConfigClasses	VARIANT-LINK-TIME: LINK
	VARIANT-POST-BUILD: POST-BUILD
	VARIANT-PRE-COMPILE: PRE-COMPILE
postBuildVariantValue	true
valueConfigClasses	VARIANT-LINK-TIME: LINK
	VARIANT-POST-BUILD: POST-BUILD
	VARIANT-PRE-COMPILE: PRE-COMPILE
requiresSymbolicNameValue	true
destination	/AUTOSAR/EcucDefs/Dem/DemConfigSet/DemEventParameter

4.136 Reference ETH_E_ALIGNMENT

Reference to the DemEventParameter which shall be issued when the error 'Alignment Error' has occurred.

Property	Value
type	ECUC-REFERENCE-DEF
origin	AUTOSAR_ECUC
lowerMultiplicity	0
upperMultiplicity	1
postBuildVariantMultiplicity	true
multiplicityConfigClasses	VARIANT-LINK-TIME: LINK
	VARIANT-POST-BUILD: POST-BUILD
	VARIANT-PRE-COMPILE: PRE-COMPILE
postBuildVariantValue	true
valueConfigClasses	VARIANT-LINK-TIME: LINK
	VARIANT-POST-BUILD: POST-BUILD
	VARIANT-PRE-COMPILE: PRE-COMPILE
requiresSymbolicNameValue	true
destination	/AUTOSAR/EcucDefs/Dem/DemConfigSet/DemEventParameter

4.137 Reference ETH_E_SINGLECOLLISION

Reference to the DemEventParameter which shall be issued when the error 'Single Collision' has occurred.

Property	Value
type	ECUC-REFERENCE-DEF
origin	AUTOSAR_ECUC
lowerMultiplicity	0
upperMultiplicity	1
postBuildVariantMultiplicity	true
multiplicityConfigClasses	VARIANT-LINK-TIME: LINK
	VARIANT-POST-BUILD: POST-BUILD
	VARIANT-PRE-COMPILE: PRE-COMPILE
postBuildVariantValue	true
valueConfigClasses	VARIANT-LINK-TIME: LINK
	VARIANT-POST-BUILD: POST-BUILD
	VARIANT-PRE-COMPILE: PRE-COMPILE
requiresSymbolicNameValue	true
destination	/AUTOSAR/EcucDefs/Dem/DemConfigSet/DemEventParameter

4.138 Reference ETH_E_MULTIPLECOLLISION

Reference to the DemEventParameter which shall be issued when the error 'Multiple Collisions' has occurred.

Property	Value
type	ECUC-REFERENCE-DEF
origin	AUTOSAR_ECUC
lowerMultiplicity	0
upperMultiplicity	1
postBuildVariantMultiplicity	true
multiplicityConfigClasses	VARIANT-LINK-TIME: LINK
	VARIANT-POST-BUILD: POST-BUILD
	VARIANT-PRE-COMPILE: PRE-COMPILE
postBuildVariantValue	true
valueConfigClasses	VARIANT-LINK-TIME: LINK
	VARIANT-POST-BUILD: POST-BUILD
	VARIANT-PRE-COMPILE: PRE-COMPILE
requiresSymbolicNameValue	true
destination	/AUTOSAR/EcucDefs/Dem/DemConfigSet/DemEventParameter

4.139 Reference ETH_E_LATECOLLISION

Reference to the DemEventParameter which shall be issued when the error 'Late Collisions' has occurred.

Property	Value
type	ECUC-REFERENCE-DEF
origin	AUTOSAR_ECUC
lowerMultiplicity	0
upperMultiplicity	1
postBuildVariantMultiplicity	true
multiplicityConfigClasses	VARIANT-LINK-TIME: LINK
	VARIANT-POST-BUILD: POST-BUILD
	VARIANT-PRE-COMPILE: PRE-COMPILE
postBuildVariantValue	true
valueConfigClasses	VARIANT-LINK-TIME: LINK
	VARIANT-POST-BUILD: POST-BUILD
	VARIANT-PRE-COMPILE: PRE-COMPILE
requiresSymbolicNameValue	true
destination	/AUTOSAR/EcucDefs/Dem/DemConfigSet/DemEventParameter

4.140 Container CommonPublishedInformation

Common container, aggregated by all modules. It contains published information about vendor and versions.

Included subcontainers:

- None

Property	Value
type	ECUC-PARAM-CONF-CONTAINER-DEF
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A

4.141 Parameter ModuleId

Module ID of this module from the BSW Module List.

Property	Value
type	ECUC-INTEGER-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-LINK-TIME: PUBLISHED-INFORMATION
	VARIANT-POST-BUILD: PUBLISHED-INFORMATION
	VARIANT-PRE-COMPILE: PUBLISHED-INFORMATION
defaultValue	88
max	88
min	88

4.142 Parameter VendorId

Vendor ID of the dedicated implementation of this module according to the AUTOSAR vendor list.

Property	Value
type	ECUC-INTEGER-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-LINK-TIME: PUBLISHED-INFORMATION
	VARIANT-POST-BUILD: PUBLISHED-INFORMATION
	VARIANT-PRE-COMPILE: PUBLISHED-INFORMATION
defaultValue	43
max	43
min	43

4.143 Parameter VendorApiInfix

In driver modules which can be instantiated several times on a single ECU,

BSW00347 requires that the name of APIs is extended by the VendorId and a vendor specific name.

This parameter is used to specify the vendor specific name.

In total, the implementation specific name is generated as follows:

[ModuleName]_[VendorId]_[VendorApiInfix][API name from SWS].

E.g. Assuming that the VendorId of the implementor is 123 and the implementer chose a

VendorApiInfix of 'v11r456' an API named Can_Write defined in the SWS will translate to Can_123_v11r456Write.

This parameter is mandatory for all modules with upper multiplicity > 1.

It shall not be used for modules with upper multiplicity = 1.

Property	Value
type	ECUC-STRING-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	0
upperMultiplicity	1
postBuildVariantMultiplicity	false
multiplicityConfigClasses	VARIANT-LINK-TIME: PUBLISHED-INFORMATION
	VARIANT-POST-BUILD: PUBLISHED-INFORMATION
	VARIANT-PRE-COMPILE: PUBLISHED-INFORMATION
postBuildVariantValue	false
valueConfigClasses	VARIANT-LINK-TIME: PUBLISHED-INFORMATION
	VARIANT-POST-BUILD: PUBLISHED-INFORMATION
	VARIANT-PRE-COMPILE: PUBLISHED-INFORMATION
defaultValue	

4.144 Parameter ArReleaseMajorVersion

Major version number of AUTOSAR specification on which the appropriate implementation is based on.

Property	Value
type	ECUC-INTEGER-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-LINK-TIME: PUBLISHED-INFORMATION
	VARIANT-POST-BUILD: PUBLISHED-INFORMATION

Property	Value
	VARIANT-PRE-COMPILE: PUBLISHED-INFORMATION
defaultValue	4
max	4
min	4

4.145 Parameter ArReleaseMinorVersion

Minor version number of AUTOSAR specification on which the appropriate implementation is based on.

Property	Value
type	ECUC-INTEGER-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-LINK-TIME: PUBLISHED-INFORMATION
	VARIANT-POST-BUILD: PUBLISHED-INFORMATION
	VARIANT-PRE-COMPILE: PUBLISHED-INFORMATION
defaultValue	4
max	4
min	4

4.146 Parameter ArReleaseRevisionVersion

Revision version number of AUTOSAR specification on which the appropriate implementation is based on.

Property	Value
type	ECUC-INTEGER-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-LINK-TIME: PUBLISHED-INFORMATION
	VARIANT-POST-BUILD: PUBLISHED-INFORMATION

Property	Value
	VARIANT-PRE-COMPILE: PUBLISHED-INFORMATION
defaultValue	0
max	0
min	0

4.147 Parameter SwMajorVersion

Major version number of the vendor specific implementation of the module. The numbering is vendor specific.

Property	Value
type	ECUC-INTEGER-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-LINK-TIME: PUBLISHED-INFORMATION
	VARIANT-POST-BUILD: PUBLISHED-INFORMATION
	VARIANT-PRE-COMPILE: PUBLISHED-INFORMATION
defaultValue	2
max	2
min	2

4.148 Parameter SwMinorVersion

Minor version number of the vendor specific implementation of the module. The numbering is vendor specific.

Property	Value
type	ECUC-INTEGER-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-LINK-TIME: PUBLISHED-INFORMATION
	VARIANT-POST-BUILD: PUBLISHED-INFORMATION

Property	Value
	VARIANT-PRE-COMPILE: PUBLISHED-INFORMATION
defaultValue	0
max	0
min	0

4.149 Parameter SwPatchVersion

Patch version number of the vendor specific implementation of the module. The numbering is vendor specific.

Property	Value
type	ECUC-INTEGER-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-LINK-TIME: PUBLISHED-INFORMATION
	VARIANT-POST-BUILD: PUBLISHED-INFORMATION
	VARIANT-PRE-COMPILE: PUBLISHED-INFORMATION
defaultValue	0
max	0
min	0

Chapter 5

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5.1 Software Specification

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6.1 Data Structures

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Chapter 7

Module Documentation

7.1 ETH Driver

7.1.1 Detailed Description

7.1.1.1 Functions Expectation called for transmission in polling mode:

- Eth_ProvideTxBuffer
- Eth_Transmit
- Eth_TxConfirmation

Function Reference

- void Eth_Init (const Eth_ConfigType *CfgPtr)
  Initializes the Ethernet Driver.
- Std_ReturnType Eth_SetControllerMode (uint8 CtrlIdx, Eth_ModeType CtrlMode)
  Enables or disables the given controller.
- Std_ReturnType Eth_GetControllerMode (uint8 CtrlIdx, Eth_ModeType *CtrlModePtr)
  Obtains the mode of the given controller.
- Std_ReturnType Eth_TxTimeAwareShaperInit (uint8 CtrlIdx)
  Control transmit frame in duration time.
- void Eth_GetPhysAddr (uint8 CtrlIdx, uint8 *PhysAddrPtr)
  Obtains the physical source address used by the indexed controller (the node MAC address).
- void Eth_SetPhysAddr (uint8 CtrlIdx, const uint8 *PhysAddrPtr)
  Set or change physical address to the defined controller.
- BufReq_ReturnType Eth_ProvideTxBuffer (uint8 CtrlIdx, uint8 Priority, Eth_BufIdxType *BufIdxPtr, uint8 **BufPtr, uint16 *LenBytePtr)
  Provides access to a transmit buffer of the specified controller.
- Std_ReturnType Eth_Transmit (uint8 CtrlIdx, Eth_BufIdxType BufIdx, Eth_FrameType FrameType, boolean TxConfirmation, uint16 LenByte, const uint8 *PhysAddrPtr)
  Triggers transmission of a previously granted and then filled transmit buffer.
- void Eth_Receive (uint8 CtrlIdx, uint8 FifoIdx, Eth_RxStatusType *RxStatusPtr)
  Triggers frames reception notifications.
- void Eth_TxConfirmation (uint8 CtrlIdx)
  Triggers frame transmission confirmations.

7.1.2 Function Reference

7.1.2.1 Eth_Init()

```
void Eth_Init (
            const Eth_ConfigType * CfgPtr )
```

Initializes the Ethernet Driver.

The configuration pointer is internally stored and the driver is initialized. The Ethernet controller is also reset.

Note

Function should be called only once.

Warning

Second call can cause undefined behavior. Call the Eth_SetControllerMode() and pass ETH_MODE_DOWN to the CtrlMode argument before the second Eth_Init call to avoid problems.

Parameters

in	*CfgPtr*	Points to the implementation specific structure containing the Eth driver configuration Compiler_Warning: this warning due to behavior of compiler depend on configs. implements Eth_Init_Activity

7.1.2.2 Eth_SetControllerMode()

```
Std_ReturnType Eth_SetControllerMode (
            uint8 CtrlIdx,
            Eth_ModeType CtrlMode )
```

Enables or disables the given controller.

Warning

Disabling the controller clears all receive and transmit buffers. The application should ensure that no data is lost.

Parameters

in	*CtrlIdx*	Index of the controller to be enabled or disabled. The index is valid within the context of the Ethernet Driver only.
in	*CtrlMode*	Mode which shall be entered • ETH_MODE_DOWN: disable the controller • ETH_MODE_ACTIVE: enable the controller

Returns

Error status

Return values

E_OK	No error was detected during the function execution.
E_NOT_OK	Development error was detected and the function failed. implements Eth_SetControllerMode_Activity

7.1.2.3 Eth_GetControllerMode()

```
Std_ReturnType Eth_GetControllerMode (
            uint8 CtrlIdx,
            Eth_ModeType * CtrlModePtr )
```

Obtains the mode of the given controller.

Parameters

in	*CtrlIdx*	Index of the controller which state shall be read. The index is valid within the context of the Ethernet Driver only.
out	*CtrlModePtr*	Pointer where to store the current controller mode.

Returns

Error status

Return values

E_OK	No error was detected during the function execution.
E_NOT_OK	Development error was detected and the function failed. implements Eth_GetControllerMode_Activity

7.1.2.4 Eth_TxTimeAwareShaperInit()

```
Std_ReturnType Eth_TxTimeAwareShaperInit (
            uint8 CtrlIdx )
```

Control transmit frame in duration time.

Parameters

in	*CtrlIdx*	Index of the controller which state shall be read. The index is valid within the context of the Ethernet Driver only.

Returns

Std_ReturnType

7.1.2.5 Eth_GetPhysAddr()

```
void Eth_GetPhysAddr (
            uint8 CtrlIdx,
            uint8 * PhysAddrPtr )
```

Obtains the physical source address used by the indexed controller (the node MAC address).

Parameters

in	*CtrlIdx*	Index of the controller whose MAC address should be read. The index is valid within the context of the Ethernet Driver only.
out	*PhysAddrPtr*	Pointer where to store the physical source address (MAC address). The address in network byte order is stored into 6 bytes at the given memory address. implements Eth_GetPhysAddr_Activity

7.1.2.6 Eth_SetPhysAddr()

```
void Eth_SetPhysAddr (
            uint8 CtrlIdx,
            const uint8 * PhysAddrPtr )
```

Set or change physical address to the defined controller.

Parameters

in	*CtrlIdx*	Index of the controller whose MAC address should be changed. The index is valid within the context of the Ethernet Driver only.
in	*PhysAddrPtr*	Pointer to the physical source address which should be set to the controller. The address is stored in 6 bytes of memory in network byte order.

Warning

This function may be called only when the controller is down. Calling function `Eth_Init` will change the controller's MAC address to the default value! implements Eth_SetPhysAddr_Activity

7.1.2.7 Eth_ProvideTxBuffer()

```
BufReq_ReturnType Eth_ProvideTxBuffer (
            uint8 CtrlIdx,
            uint8 Priority,
            Eth_BufIdxType * BufIdxPtr,
            uint8 ** BufPtr,
            uint16 * LenBytePtr )
```

Provides access to a transmit buffer of the specified controller.

Warning

The application should handle possible difference between the requested and granted buffer lengths. It is not necessary to use whole granted buffer i.e. some space at the end may not be written.

Parameters

in	*CtrlIdx*	Index of the controller which buffer shall be provided. The index is valid within the context of the Ethernet Driver only.
in	*Priority*	Frame priority for transmit buffer FIFO selection
out	*BufIdxPtr*	Index to the granted transmit buffer resource. It uniquely identifies the buffer in all subsequent calls of functions Eth_Transmit() and Eth_TxConfirmation().
out	*BufPtr*	Pointer to the granted buffer. This is the space where the data to be transmitted shall be stored.
in,out	*LenBytePtr*	Buffer payload length • In: desired length in bytes • Out: granted length in bytes

Returns

Error and buffer status

Return values

BUFREQ_OK	Buffer was successfully granted and no error has occurred.
BUFREQ_E_NOT_OK	A development error was detected and no buffer was granted.
BUFREQ_E_BUSY	All available buffers in use therefore no buffer was granted. No error has been detected. implements Eth_ProvideTxBuffer_Activity

7.1.2.8 Eth_Transmit()

```
Std_ReturnType Eth_Transmit (
            uint8 CtrlIdx,
            Eth_BufIdxType BufIdx,
            Eth_FrameType FrameType,
            boolean TxConfirmation,
            uint16 LenByte,
            const uint8 * PhysAddrPtr )
```

Triggers transmission of a previously granted and then filled transmit buffer.

Parameters

in	*CtrlIdx*	Index of the controller which buffer shall be transmitted. The index is valid within the context of the Ethernet Driver only.
in	*BufIdx*	Index of the buffer resource to be transmitted.
in	*FrameType*	Desired value of the Ethernet frame type in the frame header.
in	*TxConfirmation*	Activates transmission confirmation.
in	*LenByte*	Buffer data length in bytes (payload length).
in	*PhysAddrPtr*	Physical target address (MAC address) in network byte order.

Returns

Error status

Return values

E_OK	No error was detected during the function execution.
E_NOT_OK	Development error was detected and the function failed. implements Eth_Transmit_Activity

7.1.2.9 Eth_Receive()

```
void Eth_Receive (
            uint8 CtrlIdx,
            uint8 FifoIdx,
            Eth_RxStatusType * RxStatusPtr )
```

Triggers frames reception notifications.

All receive buffers are checked and the first received frame is passed to the EthIf module. The caller is notified whether any frame was received and whether more frames are available in the receive queue.

Parameters

in	*CtrlIdx*	Index of the controller which shall be checked whether any new frames were received. The index is valid within the context of the Ethernet Driver only.
in	*FifoIdx*	Specifies the related fifo
out	*RxStatusPtr*	Informs the caller whether a frame was received (ETH_RECEIVED or ETH_NOT_RECEIVED) and whether more frames are available in the queue (ETH_RECEIVED or ETH_RECEIVED_MORE_DATA_AVAILABLE). implements Eth_Receive_Activity

7.1.2.10 Eth_TxConfirmation()

```
void Eth_TxConfirmation (
             uint8 CtrlIdx )
```

Triggers frame transmission confirmations.

Parameters

in	*CtrlIdx*	Index of the controller which shall be checked whether any frame transmission has finished. The index is valid within the context of the Ethernet Driver only.

All transmit buffers are checked and upper layers are informed about successfully transmitted frames. Buffers containing transmitted frames are unlocked after the confirmation. implements Eth_TxConfirmation_Activity

7.2 GMAC Driver

7.2.1 Detailed Description

7.2.1.1 Platform-specific prerequisites

It is the user's responsibility to configure **DCM_GPR_DCMRWF1[RMII_MII_SEL]** to the desired PHY interface. It shall match the interface expected by the PHY and the MAC.

7.2.1.2 How to use the GMAC driver in your application

The GMAC driver does not handle clock setup or any kind of pin configuration. This is handled by the **Clock Driver** and **Pin Driver** modules, respectively. The driver assumes that the correct clock and pins configurations have been made, so it is the user's responsibility to set up clocking and pin configurations correctly.

In order to use the GMAC driver in your application, the Gmac_Ip_Init() function should be called prior to using the rest of the API. The parameters of this function specify:

- The GMAC instance to be initialized
- A structure which will hold the internal state of the driver
- A structure specifying the configuration of the GMAC module
- Two arrays of structures specifying the configuration of the buffers (one structure for each rings - an array for Rx and one for Tx)
- The MAC address to be configured for the module

The configuration of the module is specified through the Gmac_Ip_ConfigType structure and contains:

- MII-related configurations (mode, speed, duplex)
- Configuration options for the receive and transmit path
- The maximum frame length
- The MAC interrupt sources which should be enabled
- A callback function to be invoked on events
- The number of Rx and Tx rings used by the driver
- MAC address filtering options

The buffers configuration is specified through the Gmac_Ip_TxRingConfigType and Gmac_Ip_RxRingConfigType structures and contains:

- The size of the ring
- Pointers to the beginning of the buffer descriptor rings

- Pointers to the beginning of the memory area where the buffer data resides
- Channel interrupt sources
- A callback function to be invoked on per channel events
- The DMA burst length
- A list of VLAN priorities
- The queue operation mode - only for Tx rings
- The scheduling weight/quantum for DCB queues - only for Tx rings
- Idle and send slope credits for AVB queues - only for Tx rings
- High and low credit limits for AVB queues - only for Tx rings

All untagged packets will be received by ring 0 (zero). VLAN tagged packets are routed depending on the VLAN Tag Priority field and the way these priorities were assigned to the rings in the configuration.

In order to de-initialize the driver, the Gmac_Ip_Deinit() function shall be used. This function will disable the GMAC interrupts and the module, so calling other GMAC driver functions after de-initializing the driver will have undefined behavior. In order to use the driver again, Gmac_Ip_Init() should be called.

7.2.1.3 How to enable zero-copy optimization

The buffers allocation is controlled by the following configuration parameters:

- EthCtrlAllocateTxDataBuffers
- EthCtrlAllocateRxDataBuffers

Setting them both to FALSE will prevent the driver from using internal buffers.

However, it is important to keep in mind that the programming sequences will also be slightly affected:

- EthAllocateTxDataBuffers = TRUE
  - Call sequence for each frame: Gmac_Ip_GetTxBuff -> Gmac_Ip_SendFrame
  - Gmac_Ip_SendFrame must be called with the same buffer returned by Gmac_Ip_GetTxBuff
- EthAllocateTxDataBuffers = FALSE
  - Call sequence for each frame: Gmac_Ip_SendFrame
  - Gmac_Ip_SendFrame can be called with any buffer provided by the application
- EthCtrlAllocateRxDataBuffers = TRUE
  - Call sequence for each frame: Gmac_Ip_ReadFrame -> Gmac_Ip_ProvideRxBuff
  - Gmac_Ip_ProvideRxBuff must be called with the same buffer returned by Gmac_Ip_ReadFrame

- EthCtrlAllocateRxDataBuffers = FALSE

  1. Application has an RX buffer pool

     - After controller initialization: Call Gmac_Ip_ProvideRxBuff for each buffer
     - Call sequence for each frame: Gmac_Ip_ReadFrame -> Gmac_Ip_ProvideRxBuff
     - Gmac_Ip_ProvideRxBuff can be called with either:

       * The same buffer returned by Gmac_Ip_ReadFrame if the application wants to reuse the buffer
       * A buffer different than the one returned by Gmac_Ip_ReadFrame if the application doesn't want to reuse the buffer

  2. Application doesn't have an RX buffer pool and wants to provide RX buffers on demand

     - After controller initialization: Nothing to be done
     - Call sequence for each frame: Gmac_Ip_ProvideRxBuff -> Gmac_Ip_ReadFrame

7.2.1.4 How to calculate size of ethernet buffer areas

The contents of this section are relevant only when EthCtrlAllocateTxDataBuffers = TRUE and/or EthCtrlAllocate↩ RxDataBuffers = TRUE.

Ethernet buffer areas is placed in non-cacheable memory for each queue in each controller with order:

- Rx descriptor buffer areas: number of buffers configured(by EthCtrlConfigIngressFifoBufTotal) ∗ size of each buffer (Gmac_Ip_BufferDescriptorType).
- Rx data buffer areas: number of buffers configured(by EthCtrlConfigIngressFifoBufTotal) ∗ size of each buffer (EthCtrlConfigIngressFifoBufLenByte).
- Tx descriptor buffer areas: number of buffers configured(by EthCtrlConfigEgressFifoBufTotal) ∗ size of each buffer (Gmac_Ip_BufferDescriptorType).
- Tx data buffer areas: number of buffers configured(by EthCtrlConfigEgressFifoBufTotal) ∗ size of each buffer (EthCtrlConfigEgressFifoBufLenByte).

All of above memory areas need to align with start address by FEATURE_GMAC_BUFFDESCR_ALIGNMEN↩ T_BYTES(8U).

This configuration will be placed in loop with multiple queues and multiple controllers.

7.2.1.4.1 In code: - For Reception buffer descriptors for Rx Ring [!"num:i($RingIdx)"!] :

VAR_ALIGN(Gmac_Ip_BufferDescriptorType GMAC_[!"num:i($CtrlIdx)"!]_RxRing_[!"num:i($RingIdx)"!]_DescBuffer[GMA
FEATURE_GMAC_BUFFDESCR_ALIGNMENT_BYTES)

- For Reception data buffers for Rx Ring [!"num:i($RingIdx)"!]

VAR_ALIGN(uint8 GMAC_[!"num:i($CtrlIdx)"!]_RxRing_[!"num:i($RingIdx)"!]_DataBuffer[GMAC_[!"num:i($CtrlIdx)"!]_M
* GMAC_[!"num:i($CtrlIdx)"!]_MAX_RXBUFFLEN_SUPPORTED], FEATURE_GMAC_BUFFDATA_ALIGNMENT_BY

- For Transmission buffer descriptors for Tx Ring [!"num:i($RingIdx)"!]

VAR_ALIGN(Gmac_Ip_BufferDescriptorType GMAC_[!"num:i($CtrlIdx)"!]_TxRing_[!"num:i($RingIdx)"!]_DescBuffer[GMA
FEATURE_GMAC_BUFFDESCR_ALIGNMENT_BYTES)

- For Transmission data buffers for Tx Ring [!"num:i($RingIdx)"!]

VAR_ALIGN(uint8 GMAC_[!"num:i($CtrlIdx)"!]_TxRing_[!"num:i($RingIdx)"!]_DataBuffer[GMAC_[!"num:i($CtrlIdx)"!]_M
* GMAC_[!"num:i($CtrlIdx)"!]_MAX_TXBUFFLEN_SUPPORTED], FEATURE_GMAC_BUFFDATA_ALIGNMENT_BY

7.2.1.5 How to set up MAC address filters

The Address Filtering Module of the MAC checks the source address and destination address fields of each incoming packet. By default, the module only receives the frames having the destination address equal to the MAC address with which the GMAC module has been initialized or the broadcast address.

The module has 1 additional 48-bit perfect address filter with masks for each byte, which can be configured for either source address or destination address filtering. This filter can be set-up by calling the Gmac_Ip_SetAddrPerfectFilter() function.

Multicast and unicast destination addresses can be added to the hash filter by calling the Gmac_Ip_AddDstAddrToHashFilter() function and removed by calling the Gmac_Ip_RemoveDstAddrFromHashFilter() function.

The choice between perfect and hash filters can be done by calling the Gmac_Ip_SetUnicastHashFilter(), Gmac_Ip_SetMulticastHashFilter() and Gmac_Ip_SetHashOrPerfectFilter() functions.

For multicast and broadcast frames, the filters can be bypassed by calling the Gmac_Ip_SetBroadcastForwardAll() and Gmac_Ip_SetMulticastForwardAll() functions.

7.2.1.6 How to set up VLAN control and filters

Processing of IEEE802.1Q VLAN tags in both transmitted and received packets can be enabled and configured by calling the Gmac_Ip_EnableVlan() function. This function takes a parameter of type Gmac_Ip_VlanConfigType which encapsulates the following properties:

- Enables processing of up to two VLAN tags in transmitted and received packets (if present)
- Enables the transmitter and receiver to consider the S-VLAN packets (Type = 0x88A8) as valid VLAN tagged packets
- Selects if outer VLAN tag and type are deleted, inserted or replaced in transmitted packets

- Selects if inner VLAN tag and type are deleted, inserted or replaced in transmitted packets; effective only if double VLAN processing is enabled
- Selects if outer VLAN tag is stripped in received packets
- Selects if inner VLAN tag is stripped in received packets

The user has the option to configure both outer VLAN tag and type for an active transmission ring, to be automatically inserted or replaced for each packet, by calling the Gmac_Ip_SetTxOuterVlanTagForInsertion() and Gmac_Ip_SetTxOuterVlanTagForReplacement functions. Additionally, both inner VLAN tag and type can be configured for all transmission rings, by calling the Gmac_Ip_SetTxInnerVlanTag() function.

By default, all VLAN tagged packets are accepted. The Rx filter can be enabled and configured by calling the Gmac_Ip_SetVlanTagRxFilter() function. The matching options that can be configured are the following:

- Enable matching on the inner VLAN tag (if present); by default, the outer VLAN tag is compared
- Enable matching for S-VLAN (Type = 0x88A8) packets; by default, matching is performed for C-VLAN (Type = 0x8100) packets
- Disable matching for the VLAN type (only the VLAN tag is compared); by default, the VLAN type is compared
- Enable VLAN tag inverse filter match
- Enable 12-bit VLAN identifier match; by default, the complete 16-bit VLAN tag is compared

The available perfect filters (see the FEATURE_GMAC_VLAN_RX_FILTERS_NUM define for the the specific maximum number of filters configurable on each platform) can be set, read and cleared by the means of the following functions: Gmac_Ip_AddVlanTagToRxFilter(), Gmac_Ip_ReadVlanTagRxFilter(), Gmac_Ip_ClearVlanTagRx↩ Filter(). These filters are used for perfect matching and the above described matching criteria are applied to all of them, based on the user configuration. Optionally, a hash table can be used for group matching of 16-bit outer VLAN tags, irrespective of the VLAN type or other criteria. VLAN tags can be adder or removed from to/from the hash table by calling the Gmac_Ip_AddVlanTagToHashTable() or Gmac_Ip_RemoveVlanTagFromHashTable() functions.

When the Rx filters are enabled, the packets with unmatching VLAN tags will be rejected.

7.2.1.7 How to set up the System Time and enable timestamping for packets

For the timestamp to be captured for Tx and Rx packets, the System Time must be first initialized by calling the Gmac_Ip_InitSysTime() function. The timestamp is available as extended information in Gmac_Ip_TxInfoType and Gmac_Ip_RxInfoType structures, passed as parameters to Gmac_Ip_GetTransmitStatus() and Gmac_Ip_ReadFrame().

Please note that the clock source for the GMAC System Time is not handled by the GMAC driver and the user must ensure that it is correctly enabled prior to initializing the Sytem Time.

7.2.1.8 Examples:

7.2.1.8.1 Initializing the module:

```
Gmac_Ip_Init(INST_GMAC_0, &Gmac_0_ConfigPB_<functionalGroupName>);

/* ... */

Gmac_Ip_Deinit(INST_GMAC_0);
```

7.2.1.8.2 Sending a frame (EthCtrlAllocateTxDataBuffers = TRUE):

```
Gmac_Ip_BufferType buffer;
Gmac_Ip_TxOptionsType options = {FALSE, GMAC_CRC_AND_PAD_INSERTION, GMAC_CHECKSUM_INSERTION
buffer.length = 8;
if (GMAC_STATUS_SUCCESS == Gmac_Ip_GetTxBuff(INST_GMAC_0, 0, &buffer, NULL_PTR))
{
    for (uint32 i = 0; i < 8; ++i)
    {
        buffer.data[i] = i;
    }
    Gmac_Ip_SendFrame(INST_GMAC_0, 0, &buffer, &options);
}
```

7.2.1.8.3 Sending a frame (EthCtrlAllocateTxDataBuffers = FALSE):

```
Gmac_Ip_BufferType buffer;
Gmac_Ip_TxOptionsType options = {FALSE, GMAC_CRC_AND_PAD_INSERTION, GMAC_CHECKSUM_INSERTION
uint8_t data[8] = {0, 1, 2, 3, 4, 5, 6, 7};
buffer.data = data;
buffer.length = 8;
Gmac_Ip_SendFrame(INST_GMAC_0, 0, &buffer, &options);
```

7.2.1.8.4 Receiving a frame - polling method (EthCtrlEnableRxInterrupt = FALSE):

```
Gmac_Ip_BufferType buffer;
Gmac_Ip_RxInfoType info;
for (;;)
{
    if (status == Gmac_Ip_ReadFrame(INST_GMAC_0, 0, &buffer, &info))
    {
        /* Process buffer */
        /* buffer is no longer needed, provide it to the driver in order to be used by the reception mechanism */
        Gmac_Ip_ProvideRxBuff(INST_GMAC_0, 0, &buffer);
    }
}
```

7.2.1.8.5 Receiving a frame - interrupt method (EthCtrlEnableRxInterrupt = TRUE): void rx_callback(uint8 instance, uint8 channel)

```
{
    Gmac_Ip_BufferType buffer;
    if (status == Gmac_Ip_ReadFrame(INST_GMAC_0, 0, &buffer, NULL_PTR))
    {
        /* Process buffer */
        /* buffer is no longer needed, provide it to the driver in order to be  used by the reception mechanism */
        Gmac_Ip_ProvideRxBuff(INST_GMAC_0, 0, &buffer);
    }
}
int main(void)
{
    /* ... */

    Gmac_Ip_Init(INST_GMAC_0, &Gmac_0_ConfigPB_<functionalGroupName>);
    /* ... */
}
```

7.2.1.8.6 Enabling and configuring VLAN processing and filtering: Gmac_Ip_VlanConfigType gmac_0_vlanConfig = {

```
    /* .enDoubleVlan = */ TRUE,
    /* .enSvlan = */ FALSE,
    /* .outerVlanIns   = */ GMAC_VLAN_TAG_INSERTION,
    /* .innerVlanIns   = */ GMAC_VLAN_TAG_REPLACEMENT,
    /* .outerVlanStrip = */ GMAC_VLAN_TAG_DO_NOT_STRIP,
    /* .innerVlanStrip = */ GMAC_VLAN_TAG_STRIP_IF_FILTER_FAIL,
};
Gmac_Ip_VlanRxFilterType gmac_0_vlanRxFilterConfig = {
```

```
    /* .enInnerVlanMatch = */ FALSE,
    /* .enSvlanMatch = */ FALSE,
    /* .disVlanTypeMatch = */ FALSE,
    /* .enInverseMatch = */ FALSE,
    /* .en12bitMatch = */ TRUE,
};
/* Enable VLAN tag processing. */
Gmac_Ip_EnableVlan(INST_GMAC_0, &gmac_0_vlanConfig);
/* Set the outer VLAN tag to inserted for each transmitted packet (a different tag is inserted
   depending on the used transmission ring). */
Gmac_Ip_SetTxOuterVlanTagForInsertion(INST_GMAC_0, 0U, GMAC_VLAN_TYPE_C_VLAN, 0x1U);
Gmac_Ip_SetTxOuterVlanTagForInsertion(INST_GMAC_0, 1U, GMAC_VLAN_TYPE_C_VLAN, 0x2U);
Gmac_Ip_SetTxOuterVlanTagForInsertion(INST_GMAC_0, 2U, GMAC_VLAN_TYPE_C_VLAN, 0x3U);
/* Set the inner VLAN tag to be replaced for each transmitted packet. */
Gmac_Ip_SetTxOuterVlanTagForReplacement(INST_GMAC_0, GMAC_VLAN_TYPE_S_VLAN, 0x03U);
/* Enable VLAN tag Rx filters. */
Gmac_Ip_SetVlanTagRxFilter(INST_GMAC_0, TRUE, &gmac_0_vlanRxFilterConfig);
/* Add different VLAN tags to the perfect filters. */
Gmac_Ip_AddVlanTagToRxFilter(INST_GMAC_0, 0U, 0x6003U);
Gmac_Ip_AddVlanTagToRxFilter(INST_GMAC_0, 1U, 0x6001U);
Gmac_Ip_AddVlanTagToRxFilter(INST_GMAC_0, 2U, 0x2U);
/* Add VLAN tags to the hash table for group filtering. */
Gmac_Ip_AddVlanTagToHashTable(INST_GMAC_0, 0x4U);
Gmac_Ip_AddVlanTagToHashTable(INST_GMAC_0, 0x5U);
```

Data Structures

- struct Gmac_Ip_TxRingConfigType

  Specialization of ring configuration for Tx Rings. More...
- struct Gmac_Ip_RxRingConfigType

  Specialization of ring configuration for Rx Rings. More...
- struct Gmac_Ip_BufferType

  Send/Receive buffer information for the user. More...
- struct Gmac_Ip_TimestampType

  Defines the timestamp. More...
- struct Gmac_Ip_SysTimeConfigType

  Defines a structure to configure the system time. More...
- struct Gmac_Ip_TxOptionsType

  Transmit options for a particular frame. More...
- struct Gmac_Ip_TxInfoType

  Detailed status of a transmit buffer descriptor. More...
- struct Gmac_Ip_RxInfoType

  Detailed status of a receive buffer descriptor. More...
- struct Gmac_Ip_VlanConfigType

  Defines the VLAN configuration structure. More...

Macros

- #define GMAC_TX_ERR_IP_HEADER_ERROR

  When set, it indicates that the Checksum Offload engine detected an IP header error.
- #define GMAC_TX_ERR_JABBER_TIMEOUT

  When set, it indicates that the MAC transmitter has experienced a jabber time-out.
- #define GMAC_TX_ERR_PACKET_FLUSHED

  When set, it indicates that the GMAC module flushed the packet because of a software flush command given by the CPU.
- #define GMAC_TX_ERR_PAYLOAD_CHECKSUM_ERROR

  When set, it indicates that the Checksum Offload engine had a failure and did not insert any checksum into the encapsulated TCP, UDP, or ICMP payload.
- #define GMAC_TX_ERR_LOSS_OF_CARRIER

  When set, it indicates that Loss of Carrier occurred during packet transmission. This is valid only for the packets transmitted without collision and when the MAC operates in the half-duplex mode.
- #define GMAC_TX_ERR_NO_CARRIER

  When set, it indicates that the carrier sense signal form the PHY was not asserted during transmission.
- #define GMAC_TX_ERR_LATE_COLLISION

  When set, it indicates that packet transmission was aborted because a collision occurred after the collision window (64 byte times including Preamble in MII mode and 512 byte times including Preamble and Carrier Extension in GMII mode). This bit is not valid if Underflow Error is set.
- #define GMAC_TX_ERR_EXCESSIVE_COLLISION

  When set, it indicates that the transmission was aborted after 16 successive collisions while attempting to transmit the current packet.
- #define GMAC_TX_ERR_EXCESSIVE_DEFERRAL

When set, it indicates that the transmission ended because of excessive deferral of over 24,288 bit times (155,680 bits times in 1000 Mbps mode or Jumbo Packet enabled mode) if deferral check was set at init.

- #define GMAC_TX_ERR_UNDERFLOW_ERROR

  When set, it indicates that the MAC aborted the packet because the data arrived late from the system memory.
- #define GMAC_RX_ERR_CRC_ERROR

  When this bit is set, it indicates that a CRC Error occurred on the received packet.
- #define GMAC_RX_ERR_DRIBBLE_ERROR

  When this bit is set, it indicates that the received packet has a non-integer multiple of bytes (odd nibbles). This bit is valid only in the MII Mode.
- #define GMAC_RX_ERR_RECEIVE_ERROR

  When this bit is set, it indicates that the Receive error signal and Receive data valid signal are asserted during packet reception. This error also includes carrier extension error in the GMII and half-duplex mode.
- #define GMAC_RX_ERR_WATCHDOG_TIMEOUT

  When this bit is set, it indicates that the Receive Watchdog Timer has expired while receiving the current packet. The current packet is truncated after watchdog timeout.
- #define GMAC_RX_ERR_OVERFLOW_ERROR

  When this bit is set, it indicates that the received packet is damaged because of buffer overflow in Rx FIFO.
- #define GMAC_RX_ERR_GIANT_PACKET

  When this bit is set, it indicates that the packet length exceeds the specified maximum Ethernet size of 1518, 1522, or 2000 bytes (9018 or 9022 bytes if jumbo packet enable is set).

Types Reference

- typedef void(* Gmac_Ip_CallbackType) (const uint8 Instance)

  Callback function invoked when a general event is encountered.
- typedef void(* Gmac_Ip_ChCallbackType) (const uint8 Instance, const uint8 Channel)

  Callback function invoked when a channel event is encountered.

Enum Reference

- enum Gmac_Ip_StatusType

  Return values.
- enum Gmac_Ip_CounterType

  Statistics counters enumeration.
- enum Gmac_Ip_PowerStateType

  Controller power state indication.
- enum Gmac_Ip_SpeedType

  Media Independent Interface speed selection.
- enum Gmac_Ip_MacAddrFilterType

  Source/Destination address filter selection.
- enum Gmac_Ip_CrcPadControlType

  CRC and Pad insertion control.
- enum Gmac_Ip_ChecksumInsControlType

  Checksum insertion control.
- enum Gmac_Ip_PayloadType

  Type of payload encapsulated in the IP datagram.

- enum Gmac_Ip_VlanType

  VLAN type.
- enum Gmac_Ip_VlanInsControlType

  VLAN tag and type deletion, insertion, or replacement control for transmitted packets.
- enum Gmac_Ip_VlanStripControlType

  VLAN tag stripping control for received packets.
- enum Gmac_Ip_SysTimeCorrOffsetType

  System time correction offset.
- enum Gmac_Ip_TxSchedAlgoType

  Transmitter scheduling algorithm.
- enum Gmac_Ip_TxThresholdType

  Available Tx threshold values.

Initialization and De-initialization

- Gmac_Ip_StatusType Gmac_Ip_Init (uint8 Instance, const Gmac_CtrlConfigType *Config)

  Initializes the GMAC module.
- void Gmac_Ip_Deinit (uint8 Instance)

  Deinitializes the GMAC module.
- Gmac_Ip_PowerStateType Gmac_Ip_GetPowerState (uint8 Instance)

  Gets the current power state of the GMAC module.
- void Gmac_Ip_SetPowerState (uint8 Instance, Gmac_Ip_PowerStateType PowerState)

  Sets the power state of the GMAC module.
- void Gmac_Ip_EnableController (uint8 Instance)

  Enables all configured transmit and receive buffers and then enables the controller.
- Gmac_Ip_StatusType Gmac_Ip_DisableController (uint8 Instance)

  Disables the controller and resets all the configured transmit and receive buffers.
- void Gmac_Ip_SetSpeed (uint8 Instance, Gmac_Ip_SpeedType Speed)

  Sets the speed of the MII interface.
- Gmac_Ip_StatusType Gmac_Ip_GetTxBuff (uint8 Instance, uint8 Ring, Gmac_Ip_BufferType *Buff, uint16 *BuffId)

  Provides a transmit buffer to be used by the application for transmission.
- Gmac_Ip_StatusType Gmac_Ip_SendFrame (uint8 Instance, uint8 Ring, const Gmac_Ip_BufferType *Buff, const Gmac_Ip_TxOptionsType *Options)

  Sends an Ethernet frame.
- Gmac_Ip_StatusType Gmac_Ip_SendMultiBufferFrame (uint8 Instance, uint8 Ring, const Gmac_Ip_BufferType Buffers[], const Gmac_Ip_TxOptionsType *Options, uint32 NumBuffers)

  Sends an Ethernet frame composed out of multiple buffers (not necessarily contiguous)
- Gmac_Ip_StatusType Gmac_Ip_ReadFrame (uint8 Instance, uint8 Ring, Gmac_Ip_BufferType *Buff, Gmac_Ip_RxInfoType *Info)

  Reads a received Ethernet frame.
- void Gmac_Ip_ProvideRxBuff (uint8 Instance, uint8 Ring, const Gmac_Ip_BufferType *Buff)

  Provides a receive buffer to be used by the driver for reception.
- boolean Gmac_Ip_IsFrameAvailable (uint8 Instance, uint8 Ring)

  Checks if there are more frames available in the given queue.

- Gmac_Ip_StatusType Gmac_Ip_GetTransmitStatus (uint8 Instance, uint8 Ring, const Gmac_Ip_BufferType *Buff, Gmac_Ip_TxInfoType *Info)

  Checks if the transmission of a buffer is complete.
- uint32 Gmac_Ip_GetCounter (uint8 Instance, Gmac_Ip_CounterType Counter)

  Gets statistics from the specified counter.
- void Gmac_Ip_EnableMDIO (uint8 Instance, boolean MiiPreambleDisabled, uint32 ModuleClk)

  Enables the MDIO interface.
- Gmac_Ip_StatusType Gmac_Ip_MDIORead (uint8 Instance, uint8 PhyAddr, uint8 PhyReg, uint16 *Data, uint32 TimeoutMs)

  Reads the selected register of the PHY.
- Gmac_Ip_StatusType Gmac_Ip_MDIOWrite (uint8 Instance, uint8 PhyAddr, uint8 PhyReg, uint16 Data, uint32 TimeoutMs)

  Writes the selected register of the PHY.
- Gmac_Ip_StatusType Gmac_Ip_MDIOReadMMD (uint8 Instance, uint8 PhyAddr, uint8 Mmd, uint16 PhyReg, uint16 *Data, uint32 TimeoutMs)

  Reads a register of the specified MMD in a PHY device.
- Gmac_Ip_StatusType Gmac_Ip_MDIOWriteMMD (uint8 Instance, uint8 PhyAddr, uint8 Mmd, uint16 PhyReg, uint16 Data, uint32 TimeoutMs)

  Writes a register of the specified MMD in a PHY device.
- void Gmac_Ip_SetMacAddr (uint8 Instance, const uint8 *MacAddr)

  Configures the physical address of the MAC.
- void Gmac_Ip_GetMacAddr (uint8 Instance, uint8 *MacAddr)

  Gets the physical address of the MAC.
- Gmac_Ip_StatusType Gmac_Ip_TxTimeAwareShaperInit (uint8 Instance, const Gmac_CtrlConfigType *Config)

  Initialize time aware shaper.
- uint32 Gmac_Ip_GetInterruptFlags (uint8 Instance)

  Gets a mask of the common interrupt events which occurred.
- uint32 Gmac_Ip_GetChInterruptFlags (uint8 Instance, uint8 Channel)

  Gets a mask of the interrupt events associated to a channel which occurred.
- void Gmac_Ip_SetBroadcastForwardAll (uint8 Instance, boolean Enable)

  Enables/Disables forwarding of the broadcast traffic.
- void Gmac_Ip_SetMulticastForwardAll (uint8 Instance, boolean Enable)

  Enables/Disables forwarding of the multicast traffic, irrespective of the destination MAC address.
- void Gmac_Ip_SetUnicastHashFilter (uint8 Instance, boolean Enable)

  Enables/Disables hash filtering for unicast traffic.
- void Gmac_Ip_SetMulticastHashFilter (uint8 Instance, boolean Enable)

  Enables/Disables hash filtering for multicast traffic.
- void Gmac_Ip_SetHashOrPerfectFilter (uint8 Instance, boolean Enable)

  Enables/Disables either hash or perfect filters. If the destination address of an incoming packet passes either the perfect filter or the hash filter, the packet is forwarded to the application.
- void Gmac_Ip_AddDstAddrToHashFilter (uint8 Instance, const uint8 *MacAddr)

  Adds a hardware address to the hash filter. The destination address of an incoming packet is passed through CRC logic and then compared to the entries in the hash table.
- void Gmac_Ip_RemoveDstAddrFromHashFilter (uint8 Instance, const uint8 *MacAddr)

  Removes a hardware address from the hash filter. The destination address of an incoming packet is passed through CRC logic and then compared to the entries in the hash table.
- void Gmac_Ip_SetAddrPerfectFilter (uint8 Instance, const uint8 *MacAddr, const uint8 Mask, const Gmac_Ip_MacAddrFilterType AddrType, boolean Enable)

Sets the second destination/source address perfect filter.

- void Gmac_Ip_EnableVlan (uint8 Instance, const Gmac_Ip_VlanConfigType *VlanConfig)

  Enable and set VLAN control for transmitted and received packets.
- Gmac_Ip_StatusType Gmac_Ip_SetTxOuterVlanTagForInsertion (uint8 Instance, uint8 Ring, Gmac_Ip_VlanType VlanType, uint16 VlanTag)

  Sets outer VLAN type and tag to be inserted by a particular transmission ring.
- Gmac_Ip_StatusType Gmac_Ip_SetTxOuterVlanTagForReplacement (uint8 Instance, Gmac_Ip_VlanType VlanType, uint16 VlanTag)

  Sets outer VLAN tag to be replaced for all transmission rings.
- void Gmac_Ip_SetTxInnerVlanTag (uint8 Instance, Gmac_Ip_VlanType VlanType, uint16 VlanTag)

  Sets inner VLAN type and tag to be inserted/replaced on transmission.
- void Gmac_Ip_AddVlanTagToHashTable (uint8 Instance, uint16 VlanTag)

  Adds a VLAN Tag to the Hash Table filter.
- void Gmac_Ip_RemoveVlanTagFromHashTable (uint8 Instance, uint16 VlanTag)

  Removes a VLAN Tag from the hash table filter.
- void Gmac_Ip_InitSysTime (uint8 Instance, const Gmac_Ip_SysTimeConfigType *SysTimeConfig)

  Initialize system time.
- Gmac_Ip_StatusType Gmac_Ip_SetSysTimeCorr (uint8 Instance, Gmac_Ip_SysTimeCorrOffsetType Offset, uint32 SecondsUpdate, uint32 NanoSecondsUpdate)

  Set system time correction.
- Gmac_Ip_StatusType Gmac_Ip_SetRateRatioCorr (uint8 Instance, float64 RateRatio)

  Set system time correction.
- void Gmac_Ip_GetSysTime (uint8 Instance, Gmac_Ip_TimestampType *Timestamp)

  Gets the current system time.
- void Gmac_Ip_SetTxSchedAlgo (uint8 Instance, Gmac_Ip_TxSchedAlgoType SchedAlgo)

  Sets the transmitter scheduling algorithm.
- void Gmac_Ip_SetTxRingWeight (uint8 Instance, uint8 Ring, uint32 Weight)

  Sets the weight (in WRR scheduling algorithm) for a particular Tx ring.
- void Gmac_Ip_EnableTxStoreAndForward (uint8 Instance, uint8 Ring)

  Enables the store and forward feature on the transmit path.
- void Gmac_Ip_SetTxThreshold (uint8 Instance, uint8 Ring, Gmac_Ip_TxThresholdType ThresholdValue)

  Sets transmit threshold levels.
- #define **ETH_STOP_SEC_CODE**

7.2.2 Data Structure Documentation

7.2.2.1 struct Gmac_Ip_TxRingConfigType

Specialization of ring configuration for Tx Rings.

Definition at line 572 of file Gmac_Ip_Types.h.

Data Fields

Type	Name	Description
uint32	Weight	Scheduling weight/quantum for DCB queues

Data Fields

Type	Name	Description
uint32	IdleSlopeCredit	Idle slope credit for AVB queues
uint32	SendSlopeCredit	Send slope credit for AVB queues
uint32	HiCredit	High credit limit for AVB queues
sint32	LoCredit	Low credit limit for AVB queues
Gmac_Ip_BufferDescriptorType *	RingDesc	Buffer descriptor ring start address.
Gmac_Ip_ChCallbackType	Callback	Callback function for current channel.
uint8 *	Buffer	Buffer data pool start address
uint32	Interrupts	Channel interrupt sources. A logical OR of "Gmac_Ip_ChInterruptType".
uint16	BufferLen	Length of each individual buffer in a pool
uint16	RingSize	Buffer descriptors number.
uint8	PriorityMask	A logical OR of "Gmac_Ip_VlanPriorityType".
uint8	DmaBurstLength	Maximum number of beats in one DMA block data transfer.
Gmac_Ip_OperationModeType	QueueOpMode	Queue operation mode

7.2.2.2 struct Gmac_Ip_RxRingConfigType

Specialization of ring configuration for Rx Rings.

Definition at line 595 of file Gmac_Ip_Types.h.

Data Fields

- Gmac_Ip_BufferDescriptorType * RingDesc
- Gmac_Ip_ChCallbackType Callback
- uint8 * Buffer
- uint32 Interrupts
- uint16 BufferLen
- uint16 RingSize
- uint8 PriorityMask
- uint8 DmaBurstLength

7.2.2.2.1 Field Documentation

7.2.2.2.1.1 RingDesc `Gmac_Ip_BufferDescriptorType* RingDesc`

Buffer descriptor ring start address.

Definition at line 597 of file Gmac_Ip_Types.h.

7.2.2.2.1.2 Callback Gmac_Ip_ChCallbackType Callback

Callback function for current channel.

Definition at line 598 of file Gmac_Ip_Types.h.

7.2.2.2.1.3 Buffer uint8* Buffer

Buffer data pool start address

Definition at line 599 of file Gmac_Ip_Types.h.

7.2.2.2.1.4 Interrupts uint32 Interrupts

Channel interrupt sources. A logical OR of "Gmac_Ip_ChInterruptType".

Definition at line 600 of file Gmac_Ip_Types.h.

7.2.2.2.1.5 BufferLen uint16 BufferLen

Length of each individual buffer in a pool

Definition at line 601 of file Gmac_Ip_Types.h.

7.2.2.2.1.6 RingSize uint16 RingSize

Buffer descriptors number.

Definition at line 602 of file Gmac_Ip_Types.h.

7.2.2.2.1.7 PriorityMask uint8 PriorityMask

A logical OR of "Gmac_Ip_VlanPriorityType".

Definition at line 603 of file Gmac_Ip_Types.h.

7.2.2.2.1.8 DmaBurstLength uint8 DmaBurstLength

Maximum number of beats in one DMA block data transfer.

Definition at line 604 of file Gmac_Ip_Types.h.

7.2.2.3 struct Gmac_Ip_BufferType

Send/Receive buffer information for the user.

Definition at line 612 of file Gmac_Ip_Types.h.

Data Fields

- uint8 * Data
- uint16 Length

7.2.2.3.1 Field Documentation

7.2.2.3.1.1 Data uint8* Data

Pointer to the data buffer

Definition at line 614 of file Gmac_Ip_Types.h.

7.2.2.3.1.2 Length uint16 Length

Length of the data buffer

Definition at line 615 of file Gmac_Ip_Types.h.

7.2.2.4 struct Gmac_Ip_TimestampType

Defines the timestamp.

Definition at line 622 of file Gmac_Ip_Types.h.

Data Fields

- uint32 nanoseconds
- uint32 seconds
- uint16 secondsHi

7.2.2.4.1 Field Documentation

7.2.2.4.1.1 nanoseconds `uint32 nanoseconds`

Nanoseconds part of the time.

Definition at line 624 of file Gmac_Ip_Types.h.

7.2.2.4.1.2 seconds `uint32 seconds`

The 32 least significant bits of the 48 bits seconds part of the time.

Definition at line 625 of file Gmac_Ip_Types.h.

7.2.2.4.1.3 secondsHi `uint16 secondsHi`

The 16 most significant bits of the 48 bit seconds part of the time.

Definition at line 626 of file Gmac_Ip_Types.h.

7.2.2.5 struct Gmac_Ip_SysTimeConfigType

Defines a structure to configure the system time.

Definition at line 633 of file Gmac_Ip_Types.h.

Data Fields

- uint8 SubSecondInc
- uint8 SubNanoSecondsInc
- Gmac_Ip_TimestampType * InitialTimestamp

7.2.2.5.1 Field Documentation

7.2.2.5.1.1 SubSecondInc `uint8 SubSecondInc`

The value in nanoseconds to be accumulated every clock cycle.

Definition at line 635 of file Gmac_Ip_Types.h.

7.2.2.5.1.2 SubNanoSecondsInc `uint8 SubNanoSecondsInc`

The value in sub-nanoseconds to be accumulated every clock cycle, represented in nanoseconds multiplied by 2^8. For example, if the required increment is 5.3ns, then subSecondInc should be 0x05 and subNanosecondsInc should be 0x4C (0.3 ∗ 2^8).

Definition at line 636 of file Gmac_Ip_Types.h.

7.2.2.5.1.3 InitialTimestamp `Gmac_Ip_TimestampType* InitialTimestamp`

Nanoseconds part of the time.

Definition at line 639 of file Gmac_Ip_Types.h.

7.2.2.6 struct Gmac_Ip_TxOptionsType

Transmit options for a particular frame.

Definition at line 646 of file Gmac_Ip_Types.h.

Data Fields

- boolean NoInt
- Gmac_Ip_CrcPadControlType CrcPadIns
- Gmac_Ip_ChecksumInsControlType ChecksumIns

7.2.2.6.1 Field Documentation

7.2.2.6.1.1 NoInt `boolean NoInt`

Does not generate interrupt on transmit completion.

Definition at line 648 of file Gmac_Ip_Types.h.

7.2.2.6.1.2 CrcPadIns `Gmac_Ip_CrcPadControlType CrcPadIns`

Controls the CRC and pad insertion for Tx packet.

Definition at line 649 of file Gmac_Ip_Types.h.

7.2.2.6.1.3 ChecksumIns Gmac_Ip_ChecksumInsControlType ChecksumIns

Controls the IP checksum insertion for Tx packet.

Definition at line 650 of file Gmac_Ip_Types.h.

7.2.2.7 struct Gmac_Ip_TxInfoType

Detailed status of a transmit buffer descriptor.

Definition at line 657 of file Gmac_Ip_Types.h.

Data Fields

- uint32 ErrMask
- Gmac_Ip_TimestampType Timestamp

7.2.2.7.1 Field Documentation

7.2.2.7.1.1 ErrMask uint32 ErrMask

Mask of GMAC_TX_ERR_∗, representing the errors that occurred.

Definition at line 659 of file Gmac_Ip_Types.h.

7.2.2.7.1.2 Timestamp Gmac_Ip_TimestampType Timestamp

The timestamp of the transmitted packet.

Definition at line 660 of file Gmac_Ip_Types.h.

7.2.2.8 struct Gmac_Ip_RxInfoType

Detailed status of a receive buffer descriptor.

Definition at line 709 of file Gmac_Ip_Types.h.

Data Fields

- uint32 ErrMask
- uint16 PktLen
- Gmac_Ip_PayloadType PayloadType
- boolean Ipv6
- boolean Ipv4
- boolean IpPayloadErr
- boolean IpHeaderErr
- uint16 OuterVlanTag
- uint16 InnerVlanTag
- Gmac_Ip_TimestampType Timestamp

7.2.2.8.1 Field Documentation

7.2.2.8.1.1 ErrMask `uint32 ErrMask`

Mask of GMAC_RX_ERR_* , representing the errors that occurred.

Definition at line 711 of file Gmac_Ip_Types.h.

7.2.2.8.1.2 PktLen `uint16 PktLen`

Byte length of the received packet that was transferred to system memory (including CRC).

Definition at line 712 of file Gmac_Ip_Types.h.

7.2.2.8.1.3 PayloadType `Gmac_Ip_PayloadType PayloadType`

The type of payload encapsulated in the IP datagram (this field is valid only if the Checksum Offloading feature is enabled when initializing the GMAC module).

Definition at line 713 of file Gmac_Ip_Types.h.

7.2.2.8.1.4 Ipv6 `boolean Ipv6`

Indicates that the frame is an IPv6 frame (this field is valid only if the Checksum Offloading feature is enabled when initializing the GMAC module).

Definition at line 715 of file Gmac_Ip_Types.h.

7.2.2.8.1.5 Ipv4 boolean Ipv4

Indicates that the frame is an IPv4 frame (this field is valid only if the Checksum Offloading feature is enabled when initializing the GMAC module).

Definition at line 717 of file Gmac_Ip_Types.h.

7.2.2.8.1.6 IpPayloadErr boolean IpPayloadErr

Indicates that an IP payload error has occurred (this field is valid only if the Checksum Offloading feature is enabled when initializing the GMAC module).

Definition at line 719 of file Gmac_Ip_Types.h.

7.2.2.8.1.7 IpHeaderErr boolean IpHeaderErr

Indicates that an IP header error has occurred (this field is valid only if the Checksum Offloading feature is enabled when initializing the GMAC module).

Definition at line 721 of file Gmac_Ip_Types.h.

7.2.2.8.1.8 OuterVlanTag uint16 OuterVlanTag

Contains the outer VLAN tag of the received packet.

Definition at line 723 of file Gmac_Ip_Types.h.

7.2.2.8.1.9 InnerVlanTag uint16 InnerVlanTag

Contains the inner VLAN tag of the received packet. This is valid only when Double VLAN tag processing is enabled.

Definition at line 724 of file Gmac_Ip_Types.h.

7.2.2.8.1.10 Timestamp Gmac_Ip_TimestampType Timestamp

The timestamp of the received packet.

Definition at line 726 of file Gmac_Ip_Types.h.

7.2.2.9 struct Gmac_Ip_VlanConfigType

Defines the VLAN configuration structure.

Definition at line 757 of file Gmac_Ip_Types.h.

Data Fields

- boolean EnDoubleVlan
- boolean EnSvlan
- Gmac_Ip_VlanInsControlType OuterVlanIns
- Gmac_Ip_VlanInsControlType InnerVlanIns
- Gmac_Ip_VlanStripControlType OuterVlanStrip
- Gmac_Ip_VlanStripControlType InnerVlanStrip

7.2.2.9.1 Field Documentation

7.2.2.9.1.1 EnDoubleVlan `boolean EnDoubleVlan`

Enables processing of up to two VLAN tags on Tx and Rx (if present).

Definition at line 759 of file Gmac_Ip_Types.h.

7.2.2.9.1.2 EnSvlan `boolean EnSvlan`

Enables MAC transmitter and receiver to consider the S-VLAN packets (Type = 0x88A8) as valid VLAN tagged packets.

Definition at line 760 of file Gmac_Ip_Types.h.

7.2.2.9.1.3 OuterVlanIns `Gmac_Ip_VlanInsControlType OuterVlanIns`

Selects if outer VLAN tag and type are deleted, inserted or replaced in transmitted packets.

Definition at line 761 of file Gmac_Ip_Types.h.

7.2.2.9.1.4 InnerVlanIns `Gmac_Ip_VlanInsControlType InnerVlanIns`

Selects if inner VLAN tag and type are deleted, inserted or replaced in transmitted packets. Effective only if double VLAN processing is enabled.

Definition at line 762 of file Gmac_Ip_Types.h.

7.2.2.9.1.5 OuterVlanStrip Gmac_Ip_VlanStripControlType OuterVlanStrip

Selects if outer VLAN tag is stripped in received packets.

Definition at line 764 of file Gmac_Ip_Types.h.

7.2.2.9.1.6 InnerVlanStrip Gmac_Ip_VlanStripControlType InnerVlanStrip

Selects if inner VLAN tag is stripped in received packets.

Definition at line 765 of file Gmac_Ip_Types.h.

7.2.3 Macro Definition Documentation

7.2.3.1 GMAC_TX_ERR_IP_HEADER_ERROR

```
#define GMAC_TX_ERR_IP_HEADER_ERROR
```

When set, it indicates that the Checksum Offload engine detected an IP header error.

Definition at line 664 of file Gmac_Ip_Types.h.

7.2.3.2 GMAC_TX_ERR_JABBER_TIMEOUT

```
#define GMAC_TX_ERR_JABBER_TIMEOUT
```

When set, it indicates that the MAC transmitter has experienced a jabber time-out.

Definition at line 667 of file Gmac_Ip_Types.h.

7.2.3.3 GMAC_TX_ERR_PACKET_FLUSHED

```
#define GMAC_TX_ERR_PACKET_FLUSHED
```

When set, it indicates that the GMAC module flushed the packet because of a software flush command given by the CPU.

Definition at line 671 of file Gmac_Ip_Types.h.

7.2.3.4 GMAC_TX_ERR_PAYLOAD_CHECKSUM_ERROR

```
#define GMAC_TX_ERR_PAYLOAD_CHECKSUM_ERROR
```

When set, it indicates that the Checksum Offload engine had a failure and did not insert any checksum into the encapsulated TCP, UDP, or ICMP payload.

Definition at line 675 of file Gmac_Ip_Types.h.

7.2.3.5 GMAC_TX_ERR_LOSS_OF_CARRIER

```
#define GMAC_TX_ERR_LOSS_OF_CARRIER
```

When set, it indicates that Loss of Carrier occurred during packet transmission. This is valid only for the packets transmitted without collision and when the MAC operates in the half-duplex mode.

Definition at line 680 of file Gmac_Ip_Types.h.

7.2.3.6 GMAC_TX_ERR_NO_CARRIER

```
#define GMAC_TX_ERR_NO_CARRIER
```

When set, it indicates that the carrier sense signal form the PHY was not asserted during transmission.

Definition at line 684 of file Gmac_Ip_Types.h.

7.2.3.7 GMAC_TX_ERR_LATE_COLLISION

```
#define GMAC_TX_ERR_LATE_COLLISION
```

When set, it indicates that packet transmission was aborted because a collision occurred after the collision window (64 byte times including Preamble in MII mode and 512 byte times including Preamble and Carrier Extension in GMII mode). This bit is not valid if Underflow Error is set.

Definition at line 690 of file Gmac_Ip_Types.h.

7.2.3.8 GMAC_TX_ERR_EXCESSIVE_COLLISION

```
#define GMAC_TX_ERR_EXCESSIVE_COLLISION
```

When set, it indicates that the transmission was aborted after 16 successive collisions while attempting to transmit the current packet.

Definition at line 694 of file Gmac_Ip_Types.h.

7.2.3.9 GMAC_TX_ERR_EXCESSIVE_DEFERRAL

```
#define GMAC_TX_ERR_EXCESSIVE_DEFERRAL
```

When set, it indicates that the transmission ended because of excessive deferral of over 24,288 bit times (155,680 bits times in 1000 Mbps mode or Jumbo Packet enabled mode) if deferral check was set at init.

Definition at line 699 of file Gmac_Ip_Types.h.

7.2.3.10 GMAC_TX_ERR_UNDERFLOW_ERROR

```
#define GMAC_TX_ERR_UNDERFLOW_ERROR
```

When set, it indicates that the MAC aborted the packet because the data arrived late from the system memory.

Definition at line 703 of file Gmac_Ip_Types.h.

7.2.3.11 GMAC_RX_ERR_CRC_ERROR

```
#define GMAC_RX_ERR_CRC_ERROR
```

When this bit is set, it indicates that a CRC Error occurred on the received packet.

Definition at line 730 of file Gmac_Ip_Types.h.

7.2.3.12 GMAC_RX_ERR_DRIBBLE_ERROR

```
#define GMAC_RX_ERR_DRIBBLE_ERROR
```

When this bit is set, it indicates that the received packet has a non-integer multiple of bytes (odd nibbles). This bit is valid only in the MII Mode.

Definition at line 734 of file Gmac_Ip_Types.h.

7.2.3.13 GMAC_RX_ERR_RECEIVE_ERROR

```
#define GMAC_RX_ERR_RECEIVE_ERROR
```

When this bit is set, it indicates that the Receive error signal and Receive data valid signal are asserted during packet reception. This error also includes carrier extension error in the GMII and half-duplex mode.

Definition at line 739 of file Gmac_Ip_Types.h.

7.2.3.14 GMAC_RX_ERR_WATCHDOG_TIMEOUT

```
#define GMAC_RX_ERR_WATCHDOG_TIMEOUT
```

When this bit is set, it indicates that the Receive Watchdog Timer has expired while receiving the current packet. The current packet is truncated after watchdog timeout.

Definition at line 743 of file Gmac_Ip_Types.h.

7.2.3.15 GMAC_RX_ERR_OVERFLOW_ERROR

```
#define GMAC_RX_ERR_OVERFLOW_ERROR
```

When this bit is set, it indicates that the received packet is damaged because of buffer overflow in Rx FIFO.

Definition at line 747 of file Gmac_Ip_Types.h.

7.2.3.16 GMAC_RX_ERR_GIANT_PACKET

```
#define GMAC_RX_ERR_GIANT_PACKET
```

When this bit is set, it indicates that the packet length exceeds the specified maximum Ethernet size of 1518, 1522, or 2000 bytes (9018 or 9022 bytes if jumbo packet enable is set).

Definition at line 751 of file Gmac_Ip_Types.h.

7.2.4 Types Reference

7.2.4.1 Gmac_Ip_CallbackType

```
typedef void(* Gmac_Ip_CallbackType) (const uint8 Instance)
```

Callback function invoked when a general event is encountered.

Definition at line 514 of file Gmac_Ip_Types.h.

7.2.4.2 Gmac_Ip_ChCallbackType

```
typedef void(* Gmac_Ip_ChCallbackType) (const uint8 Instance, const uint8 Channel)
```

Callback function invoked when a channel event is encountered.

Definition at line 519 of file Gmac_Ip_Types.h.

7.2.5 Enum Reference

7.2.5.1 Gmac_Ip_StatusType

```
enum Gmac_Ip_StatusType
```

Return values.

Enumerator

GMAC_STATUS_SUCCESS	Generic operation success status
GMAC_STATUS_ERROR	Generic operation failure status
GMAC_STATUS_BUSY	Generic operation busy status
GMAC_STATUS_TIMEOUT	Generic operation timeout status
GMAC_STATUS_UNSUPPORTED	Generic operation unsupported status
GMAC_STATUS_RX_QUEUE_EMPTY	There is no available frame in the receive queue
GMAC_STATUS_TX_QUEUE_FULL	There is no available space for the frame in the transmit queue
GMAC_STATUS_BUFF_NOT_FOUND	The specified buffer was not found in the queue
GMAC_STATUS_TX_BUFF_BUSY	All internal TX buffers are currently in use
GMAC_STATUS_TX_BUFF_OVERFLOW	The requested TX buffer length cannot

Definition at line 100 of file Gmac_Ip_Types.h.

7.2.5.2 Gmac_Ip_CounterType

enum Gmac_Ip_CounterType

Statistics counters enumeration.

Definition at line 236 of file Gmac_Ip_Types.h.

7.2.5.3 Gmac_Ip_PowerStateType

enum Gmac_Ip_PowerStateType

Controller power state indication.

Definition at line 297 of file Gmac_Ip_Types.h.

7.2.5.4 Gmac_Ip_SpeedType

enum Gmac_Ip_SpeedType

Media Independent Interface speed selection.

Enumerator

GMAC_SPEED_1G	Speed 1 Gbps.
GMAC_SPEED_10M	Speed 10 Mbps.
GMAC_SPEED_100M	Speed 100 Mbps.

Definition at line 309 of file Gmac_Ip_Types.h.

7.2.5.5 Gmac_Ip_MacAddrFilterType

enum Gmac_Ip_MacAddrFilterType

Source/Destination address filter selection.

Enumerator

GMAC_DST_ADRR_FILTER	Destination address filter.
GMAC_SRC_ADDR_FILTER	Source address filter.
GMAC_SRC_ADDR_INV_FILTER	Source address inverse filter.

Definition at line 356 of file Gmac_Ip_Types.h.

7.2.5.6 Gmac_Ip_CrcPadControlType

enum Gmac_Ip_CrcPadControlType

CRC and Pad insertion control.

Enumerator

GMAC_CRC_AND_PAD_INSERTION	Appends the CRC at the end of the transmitted packet of length greater than or equal to 60 bytes. Appends padding and CRC to a packet with length less than 60 bytes.
GMAC_CRC_INSERTION	Appends the CRC at the end of the transmitted packet but it does not append padding. The application should ensure that the padding bytes are present in the packet being transferred.
GMAC_CRC_INSERTION_DISABLE	Does not append the CRC at the end of the transmitted packet. The application should ensure that the padding and CRC bytes are present in the packet being transferred.
GMAC_CRC_REPLACEMENT	Replaces the last four bytes of the transmitted packet with recalculated CRC bytes. The application should ensure that the padding and CRC bytes are present in the packet being transferred.

Definition at line 367 of file Gmac_Ip_Types.h.

7.2.5.7 Gmac_Ip_ChecksumInsControlType

enum Gmac_Ip_ChecksumInsControlType

Checksum insertion control.

Enumerator

GMAC_CHECKSUM_INSERTION_DISABLE	Checksum insertion is disabled.
GMAC_CHECKSUM_INSERTION_IP	IP header checksum calculation and insertion are enabled.
GMAC_CHECKSUM_INSERTION_PROTO_NO_PSEUDOH	Protocol checksum calculation and insertion are enabled, but pseudo-header checksum is not calculated in hardware.
GMAC_CHECKSUM_INSERTION_PROTO_PSEUDOH	Protocol checksum calculation and insertion are enabled, and pseudo-header checksum is calculated in hardware.

Definition at line 383 of file Gmac_Ip_Types.h.

7.2.5.8 Gmac_Ip_PayloadType

enum Gmac_Ip_PayloadType

Type of payload encapsulated in the IP datagram.

Enumerator

GMAC_IP_PAYLOAD_TYPE_UNKNOWN	Unknown type or IP payload not processed.
GMAC_IP_PAYLOAD_TYPE_UDP	UDP packet.
GMAC_IP_PAYLOAD_TYPE_TCP	TCP packet.
GMAC_IP_PAYLOAD_TYPE_ICMP	ICMP packet.

Definition at line 397 of file Gmac_Ip_Types.h.

7.2.5.9 Gmac_Ip_VlanType

enum Gmac_Ip_VlanType

VLAN type.

Enumerator

GMAC_VLAN_TYPE_C_VLAN	C-VLAN type (0x8100).
GMAC_VLAN_TYPE_S_VLAN	S-VLAN type (0x88A8).

Definition at line 409 of file Gmac_Ip_Types.h.

7.2.5.10 Gmac_Ip_VlanInsControlType

enum Gmac_Ip_VlanInsControlType

VLAN tag and type deletion, insertion, or replacement control for transmitted packets.

Enumerator

GMAC_VLAN_TAG_NO_CONTROL	The VLAN tag is not altered in the packet to be transmitted.

Enumerator

GMAC_VLAN_TAG_DELETION	The VLAN tag is deleted from the packet to be transmitted.
GMAC_VLAN_TAG_INSERTION	The VLAN tag is inserted in the packet to be transmitted.
GMAC_VLAN_TAG_REPLACEMENT	The VLAN tag is replaced in the packet to be transmitted.

Definition at line 419 of file Gmac_Ip_Types.h.

7.2.5.11 Gmac_Ip_VlanStripControlType

enum Gmac_Ip_VlanStripControlType

VLAN tag stripping control for received packets.

Enumerator

GMAC_VLAN_TAG_DO_NOT_STRIP	Does not strip the inner VLAN tag in the received packet.
GMAC_VLAN_TAG_STRIP_IF_FILTER_PASS	Strips the inner VLAN tag in the received packet if the VLAN filter passes.
GMAC_VLAN_TAG_STRIP_IF_FILTER_FAIL	Strips the inner VLAN tag in the received packet if the VLAN filter fails.
GMAC_VLAN_TAG_ALWAYS_STRIP	Always strips the inner VLAN tag in the received packet.

Definition at line 431 of file Gmac_Ip_Types.h.

7.2.5.12 Gmac_Ip_SysTimeCorrOffsetType

enum Gmac_Ip_SysTimeCorrOffsetType

System time correction offset.

Enumerator

GMAC_SYS_TIME_CORR_POSITIVE	Correction value is added to the system time.
GMAC_SYS_TIME_CORR_NEGATIVE	Correction value is subtracted from the system time.

Definition at line 460 of file Gmac_Ip_Types.h.

7.2.5.13 Gmac_Ip_TxSchedAlgoType

```
enum Gmac_Ip_TxSchedAlgoType
```

Transmitter scheduling algorithm.

Enumerator

GMAC_SCHED_ALGO_WRR	Weighted round robin algorithm.
GMAC_SCHED_ALGO_WFQ	Weighted fair queueing algorithm.
GMAC_SCHED_ALGO_DWRR	Deficit weighted round robin algorithm.
GMAC_SCHED_ALGO_SP	Strict priority algorithm.

Definition at line 470 of file Gmac_Ip_Types.h.

7.2.5.14 Gmac_Ip_TxThresholdType

```
enum Gmac_Ip_TxThresholdType
```

Available Tx threshold values.

Definition at line 481 of file Gmac_Ip_Types.h.

7.2.6 Function Reference

7.2.6.1 Gmac_Ip_Init()

```
Gmac_Ip_StatusType Gmac_Ip_Init (
            uint8 Instance,
            const Gmac_CtrlConfigType * Config )
```

Initializes the GMAC module.

This function initializes and enables the GMAC module, configuring receive and transmit control settings, the receive and transmit descriptors rings, and the MAC physical address.

Note: All untagged packets will be received by ring 0 (zero). VLAN tagged packets are routed depending on the VLAN Tag Priority field according to the provided configuration.

Parameters

in	*instance*	Instance number
in	*config*	Pointer to the module configuration structure

Return values

GMAC_STATUS_SUCCESS	The initialization was successful.
GMAC_STATUS_TIMEOUT	The DMA subsystem reset could not complete.

7.2.6.2 Gmac_Ip_Deinit()

```
void Gmac_Ip_Deinit (
            uint8 Instance )
```

Deinitializes the GMAC module.

This function disables the interrupts and then disables the GMAC module.

Parameters

in	*instance*	Instance number

7.2.6.3 Gmac_Ip_GetPowerState()

```
Gmac_Ip_PowerStateType Gmac_Ip_GetPowerState (
            uint8 Instance )
```

Gets the current power state of the GMAC module.

Parameters

in	*instance*	Instance number

Returns

Gmac_Ip_PowerStateType The power state of the controller

7.2.6.4 Gmac_Ip_SetPowerState()

```
void Gmac_Ip_SetPowerState (
            uint8 Instance,
            Gmac_Ip_PowerStateType PowerState )
```

Sets the power state of the GMAC module.

Parameters

in	*instance*	Instance number
in	*powerState*	Power state to transition into

7.2.6.5 Gmac_Ip_EnableController()

```
void Gmac_Ip_EnableController (
            uint8 Instance )
```

Enables all configured transmit and receive buffers and then enables the controller.

Parameters

in	*instance*	Instance number

7.2.6.6 Gmac_Ip_DisableController()

```
Gmac_Ip_StatusType Gmac_Ip_DisableController (
            uint8 Instance )
```

Disables the controller and resets all the configured transmit and receive buffers.

Warning: This function ignores all pending transmission and reception requests

Parameters

in	*instance*	Instance number

Return values

GMAC_STATUS_SUCCESS	Tthe controller was successfully disabled.
GMAC_STATUS_TIMEOUT	Tthe underlying MTL queues could not be flushed.

7.2.6.7 Gmac_Ip_SetSpeed()

```
void Gmac_Ip_SetSpeed (
            uint8 Instance,
            Gmac_Ip_SpeedType Speed )
```

Sets the speed of the MII interface.

Parameters

in	*instance*	Instance number
in	*speed*	MII speed

7.2.6.8 Gmac_Ip_GetTxBuff()

```
Gmac_Ip_StatusType Gmac_Ip_GetTxBuff (
            uint8 Instance,
            uint8 Ring,
            Gmac_Ip_BufferType * Buff,
            uint16 * BuffId )
```

Provides a transmit buffer to be used by the application for transmission.

This function provides an internal buffer which can further be used by the application to store the transmit data.

Note: The buffer will be marked as locked and won't be released until after a call to Gmac_Ip_GetTransmitStatus for the same buffer returns GMAC_STATUS_SUCCESS.

Important: The driver does not ensure synchronization between different threads trying to get a buffer at the same time. This synchronization shall be implemented by the application.

Parameters

in	*instance*	Instance number
in	*ring*	Ring number
in,out	*buff*	In: Buffer containing the desired length Out: Buffer containing the granted length or available length in case of overflow.
out	*buffId*	Index of the buffer (descriptor) within the ring. If this information is not needed, this parameter should be NULL_PTR.

Return values

GMAC_STATUS_SUCCESS	The buffer has been succesfully locked.
GMAC_STATUS_TX_BUFF_BUSY	All buffers are currently in use.
GMAC_STATUS_TX_BUFF_OVERFLOW	The requested buffer length cannot be granted.

7.2.6.9 Gmac_Ip_SendFrame()

```
Gmac_Ip_StatusType Gmac_Ip_SendFrame (
```

```
            uint8 Instance,
            uint8 Ring,
            const Gmac_Ip_BufferType * Buff,
            const Gmac_Ip_TxOptionsType * Options )
```

Sends an Ethernet frame.

This function sends an Ethernet frame, contained in the buffer received as parameter.

Note: Since the transmission of the frame is not complete when this function returns, the application must not change/alter/re-use the provided buffer until after a call to Gmac_Ip_GetTransmitStatus for the same buffer returns GMAC_STATUS_SUCCESS.

Important: The driver does not ensure synchronization between different threads trying to send a frame at the same time. This synchronization shall be implemented by the application.

Parameters

in	*instance*	Instance number
in	*ring*	The ring number
in	*buff*	The buffer containing the frame
in	*options*	Configuration options applicable to this buffer's transmission only. Can be NULL_PTR, if no special option is required.

Return values

GMAC_STATUS_SUCCESS	The frame was successfully enqueued for transmission.
GMAC_STATUS_TX_QUEUE_FULL	There is no available space for the frame in the queue.

7.2.6.10 Gmac_Ip_SendMultiBufferFrame()

```
Gmac_Ip_StatusType Gmac_Ip_SendMultiBufferFrame (
            uint8 Instance,
            uint8 Ring,
            const Gmac_Ip_BufferType Buffers[],
            const Gmac_Ip_TxOptionsType * Options,
            uint32 NumBuffers )
```

Sends an Ethernet frame composed out of multiple buffers (not necessarily contiguous)

This function sends an Ethernet frame contained in the buffers received as parameter.

Note: Since the transmission of the frame is not complete when this function returns, the application must not change/alter/re-use the provided buffers until after a call to Gmac_Ip_GetTransmitStatus for the same buffer returns GMAC_STATUS_SUCCESS.

Important: The driver does not ensure synchronization between different threads trying to send a frame at the same time. This synchronization shall be implemented by the application.

Parameters

in	*Instance*	Instance number
in	*Ring*	The ring number
in	*Buffers*	The contiguous buffer array composing the frame
in	*Options*	Configuration options applied to the transmitted frame. Must be different from NULL_PTR.
in	*NumBuffers*	The size of the Buffers array

Return values

GMAC_STATUS_SUCCESS	The frame was successfully enqueued for transmission.
GMAC_STATUS_TX_QUEUE_FULL	There is no available space for the frame in the queue.

7.2.6.11 Gmac_Ip_ReadFrame()

```
Gmac_Ip_StatusType Gmac_Ip_ReadFrame (
            uint8 Instance,
            uint8 Ring,
            Gmac_Ip_BufferType * Buff,
            Gmac_Ip_RxInfoType * Info )
```

Reads a received Ethernet frame.

This function reads the first received Ethernet frame in the Rx queue. The buffer received as parameter will be updated by the driver and the .data field will point to a memory area containing the frame data.

Note: Once the application finished processing the buffer, it could be reused by the driver for further receptions by invoking Gmac_Ip_ProvideRxBuff.

Important: The driver does not ensure synchronization between different threads trying to read a frame at the same time. This synchronization shall be implemented by the application.

Parameters

in	*instance*	Instance number
in	*ring*	The ring number
out	*buff*	The buffer containing the frame
out	*info*	Enhanced information related to the data contained by this receive buffer. If this information is not needed, this parameter should be NULL_PTR.

Return values

GMAC_STATUS_SUCCESS	A frame was successfully read.
GMAC_STATUS_RX_QUEUE_EMPTY	There is no available frame in the queue.

7.2.6.12 Gmac_Ip_ProvideRxBuff()

```
void Gmac_Ip_ProvideRxBuff (
            uint8 Instance,
            uint8 Ring,
            const Gmac_Ip_BufferType * Buff )
```

Provides a receive buffer to be used by the driver for reception.

This function provides a buffer which can further be used by the reception mechanism in order to store the received data.

Note: The application can either provide a buffer previously obtained in a Gmac_Ip_ReadFrame call (when it is no longer needed after being fully processed), or allocate a new buffer. The former approach is recommended as it has a simpler usage model and re-uses the same initial memory range for the entire driver lifetime operation. The later approach could provide more flexibility, but since it involves constant memory free/alloc operations it is only recommended with an efficient pool-based memory allocator.

Important: The driver does not ensure synchronization between different threads trying to provide a buffer at the same time. This synchronization shall be implemented by the application.

Important: The application is responsible for providing one Rx buffer for every frame it receives, otherwise the reception ring can fill-up, affecting further reception.

Usage example:

stat = Gmac_Ip_ReadFrame(INST_GMAC1, 0U, &rxBuff);

if (stat == GMAC_STATUS_SUCCESS) { process_buffer(&rxBuff); Gmac_Ip_ProvideRxBuff(INST_GMAC1, 0U, &rxBuff); }

Parameters

in	*instance*	Instance number
in	*ring*	The ring number
in	*buff*	The buffer to be added to the reception ring

7.2.6.13 Gmac_Ip_IsFrameAvailable()

```
boolean Gmac_Ip_IsFrameAvailable (
            uint8 Instance,
            uint8 Ring )
```

Checks if there are more frames available in the given queue.

This function takes a peek at the given Rx queue to check if there are more Ethernet frames to be received. Its intended usage is to provide this information without also extracting the frame as "Gmac_Ip_ReadFrame".

Parameters

in	*instance*	Instance number
in	*ring*	The ring number

Return values

TRUE	There is an available frame in the queue.
FALSE	There is no available frame in the queue.

7.2.6.14 Gmac_Ip_GetTransmitStatus()

```
Gmac_Ip_StatusType Gmac_Ip_GetTransmitStatus (
            uint8 Instance,
            uint8 Ring,
            const Gmac_Ip_BufferType * Buff,
            Gmac_Ip_TxInfoType * Info )
```

Checks if the transmission of a buffer is complete.

This function checks if the transmission of the given buffer is complete.

Note: If the buffer is not found in the Tx ring, the function will return GMAC_STATUS_BUFF_NOT_FOUND.

Parameters

in	*instance*	Instance number
in	*ring*	The ring number
in	*buff*	The transmit buffer for which the status shall be checked
out	*info*	Extended information related to the buffer. If this information is not needed, this parameter should be NULL_PTR.

Return values

GMAC_STATUS_BUSY	The frame is still enqueued for transmission.
GMAC_STATUS_BUFF_NOT_FOUND	The buffer was not found in the Tx queue.
GMAC_STATUS_SUCCESS	Otherwise.

7.2.6.15 Gmac_Ip_GetCounter()

```
uint32 Gmac_Ip_GetCounter (
```

```
            uint8 Instance,
            Gmac_Ip_CounterType Counter )
```

Gets statistics from the specified counter.

Parameters

in	*instance*	Instance number
in	*counter*	The counter to be read

Returns

The value of the requested counter

7.2.6.16 Gmac_Ip_EnableMDIO()

```
void Gmac_Ip_EnableMDIO (
            uint8 Instance,
            boolean MiiPreambleDisabled,
            uint32 ModuleClk )
```

Enables the MDIO interface.

Parameters

in	*instance*	Instance number
in	*miiPreambleDisabled*	Enables/disables prepending a preamble to the MII management frame.
in	*clock*	provide module

7.2.6.17 Gmac_Ip_MDIORead()

```
Gmac_Ip_StatusType Gmac_Ip_MDIORead (
            uint8 Instance,
            uint8 PhyAddr,
            uint8 PhyReg,
            uint16 * Data,
            uint32 TimeoutMs )
```

Reads the selected register of the PHY.

Parameters

in	*instance*	Instance number

Parameters

in	*phyAddr*	PHY device address
in	*phyReg*	PHY register address
out	*data*	Data read from the PHY
in	*timeoutMs*	Timeout for the read operation (in milliseconds)

Return values

GMAC_STATUS_SUCCESS	The operation completed successfully.
GMAC_STATUS_TIMEOUT	The specified timeout expired before completing the operation.

7.2.6.18 Gmac_Ip_MDIOWrite()

```
Gmac_Ip_StatusType Gmac_Ip_MDIOWrite (
            uint8 Instance,
            uint8 PhyAddr,
            uint8 PhyReg,
            uint16 Data,
            uint32 TimeoutMs )
```

Writes the selected register of the PHY.

Parameters

in	*instance*	Instance number
in	*phyAddr*	PHY device address
in	*phyReg*	PHY register address
in	*data*	Data to be written in the specified register of the PHY
in	*timeoutMs*	Timeout for the write operation (in milliseconds)

Return values

GMAC_STATUS_SUCCESS	The operation completed successfully.
GMAC_STATUS_TIMEOUT	The specified timeout expired before completing the operation.

7.2.6.19 Gmac_Ip_MDIOReadMMD()

```
Gmac_Ip_StatusType Gmac_Ip_MDIOReadMMD (
            uint8 Instance,
```

```
            uint8 PhyAddr,
            uint8 Mmd,
            uint16 PhyReg,
            uint16 * Data,
            uint32 TimeoutMs )
```

Reads a register of the specified MMD in a PHY device.

Parameters

in	*instance*	Instance number
in	*phyAddr*	PHY device address
in	*mmd*	The MMD index of the target register
in	*phyReg*	PHY register address
out	*data*	Data read from the PHY
in	*timeoutMs*	Timeout for the read operation (in milliseconds)

Return values

GMAC_STATUS_SUCCESS	The operation completed successfully.
GMAC_STATUS_TIMEOUT	The specified timeout expired before completing the operation.

7.2.6.20 Gmac_Ip_MDIOWriteMMD()

```
Gmac_Ip_StatusType Gmac_Ip_MDIOWriteMMD (
            uint8 Instance,
            uint8 PhyAddr,
            uint8 Mmd,
            uint16 PhyReg,
            uint16 Data,
            uint32 TimeoutMs )
```

Writes a register of the specified MMD in a PHY device.

Parameters

in	*instance*	Instance number
in	*phyAddr*	PHY device address
in	*mmd*	The MMD index of the target register
in	*phyReg*	PHY register address
in	*data*	Data to be written in the specified register of the PHY
in	*timeoutMs*	Timeout for the write operation (in milliseconds)

Return values

GMAC_STATUS_SUCCESS	The operation completed successfully.
GMAC_STATUS_TIMEOUT	The specified timeout expired before completing the operation.

7.2.6.21 Gmac_Ip_SetMacAddr()

```
void Gmac_Ip_SetMacAddr (
             uint8 Instance,
             const uint8 * MacAddr )
```

Configures the physical address of the MAC.

Parameters

in	*instance*	Instance number
in	*macAddr*	The MAC address to be configured

7.2.6.22 Gmac_Ip_GetMacAddr()

```
void Gmac_Ip_GetMacAddr (
             uint8 Instance,
             uint8 * MacAddr )
```

Gets the physical address of the MAC.

Parameters

in	*instance*	Instance number
out	*macAddr*	The physical address of the MAC

7.2.6.23 Gmac_Ip_TxTimeAwareShaperInit()

```
Gmac_Ip_StatusType Gmac_Ip_TxTimeAwareShaperInit (
             uint8 Instance,
             const Gmac_CtrlConfigType * Config )
```

Initialize time aware shaper.

Parameters

in	*instance*	Instance number
in	*Parameters*	of controller will be used. return GMAC_STATUS_SUCCESS write/read to Gate control registers successfully return GMAC_STATUS_TIMEOUT write/read to Gate control registers failed

7.2.6.24 Gmac_Ip_GetInterruptFlags()

```
uint32 Gmac_Ip_GetInterruptFlags (
            uint8 Instance )
```

Gets a mask of the common interrupt events which occurred.

Parameters

in	*instance*	Instance number

Returns

A logical OR of "Gmac_Ip_InterruptType"

7.2.6.25 Gmac_Ip_GetChInterruptFlags()

```
uint32 Gmac_Ip_GetChInterruptFlags (
            uint8 Instance,
            uint8 Channel )
```

Gets a mask of the interrupt events associated to a channel which occurred.

Parameters

in	*instance*	Instance number
in	*channel*	Channel number

Returns

A logical OR of "Gmac_Ip_ChInterruptType"

7.2.6.26 Gmac_Ip_SetBroadcastForwardAll()

```
void Gmac_Ip_SetBroadcastForwardAll (
            uint8 Instance,
            boolean Enable )
```

Enables/Disables forwarding of the broadcast traffic.

Parameters

in	*instance*	Instance number
in	*enable*	If true, the application will receive all the broadcast traffic; if false, it stops forwarding this kind of traffic.

7.2.6.27 Gmac_Ip_SetMulticastForwardAll()

```
void Gmac_Ip_SetMulticastForwardAll (
            uint8 Instance,
            boolean Enable )
```

Enables/Disables forwarding of the multicast traffic, irrespective of the destination MAC address.

Parameters

in	*instance*	Instance number
in	*enable*	If true, the application will receive all the multicast traffic; if false, stops forwarding this kind of traffic.

7.2.6.28 Gmac_Ip_SetUnicastHashFilter()

```
void Gmac_Ip_SetUnicastHashFilter (
            uint8 Instance,
            boolean Enable )
```

Enables/Disables hash filtering for unicast traffic.

Note: When the hash filter is disabled, the addresses are passed through the perfect filter. By default, the perfect filter contains the hardware address configured when the MAC module is initialized. This address is always used for destination address filtering. Optionally, a second filter can be enabled by the means of Gmac_Ip_SetAddrPerfectFilter() function.

Parameters

in	*instance*	Instance number
in	*enable*	If true, the hash filtering for unicast destination addresses is enabled; if false, the hash filtering for such traffic is disabled.

7.2.6.29 Gmac_Ip_SetMulticastHashFilter()

```
void Gmac_Ip_SetMulticastHashFilter (
            uint8 Instance,
            boolean Enable )
```

Enables/Disables hash filtering for multicast traffic.

Note: When the hash filter is disabled, the addresses are passed through the perfect filter. By default, the perfect filter contains the hardware address configured when the MAC module is initialized. This address is always used for destination address filtering. Optionally, a second filter can be enabled by the means of Gmac_Ip_SetAddrPerfectFilter() function.

Parameters

in	*instance*	Instance number
in	*enable*	If true, the hash filtering for multicast destination addresses is enabled; if false, the hash filtering for such traffic is disabled.

7.2.6.30 Gmac_Ip_SetHashOrPerfectFilter()

```
void Gmac_Ip_SetHashOrPerfectFilter (
            uint8 Instance,
            boolean Enable )
```

Enables/Disables either hash or perfect filters. If the destination address of an incoming packet passes either the perfect filter or the hash filter, the packet is forwarded to the application.

Note: When the hash filter is disabled, the addresses are passed through the perfect filter. By default, the perfect filter contains the hardware address configured when the MAC module is initialized. This address is always used for destination address filtering. Optionally, a second filter can be enabled by the means of Gmac_Ip_SetAddrPerfectFilter() function.

Parameters

in	*instance*	Instance number
in	*enable*	If true, the hash filtering and perfect filtering for destination addresses are enabled; if false, only one filtering option is enabled, depending on unicast/multicast hash filter settings.

7.2.6.31 Gmac_Ip_AddDstAddrToHashFilter()

```
void Gmac_Ip_AddDstAddrToHashFilter (
            uint8 Instance,
            const uint8 * MacAddr )
```

Adds a hardware address to the hash filter. The destination address of an incoming packet is passed through CRC logic and then compared to the entries in the hash table.

Parameters

in	*instance*	Instance number
in	*macAddr*	The physical address

7.2.6.32 Gmac_Ip_RemoveDstAddrFromHashFilter()

```
void Gmac_Ip_RemoveDstAddrFromHashFilter (
            uint8 Instance,
            const uint8 * MacAddr )
```

Removes a hardware address from the hash filter. The destination address of an incoming packet is passed through CRC logic and then compared to the entries in the hash table.

Parameters

in	*instance*	Instance number
in	*macAddr*	The physical address

7.2.6.33 Gmac_Ip_SetAddrPerfectFilter()

```
void Gmac_Ip_SetAddrPerfectFilter (
            uint8 Instance,
            const uint8 * MacAddr,
            const uint8 Mask,
            const Gmac_Ip_MacAddrFilterType AddrType,
            boolean Enable )
```

Sets the second destination/source address perfect filter.

Note: By default, the perfect filter contains the hardware address configured when the MAC module is initialized. This address is always used for destination address filtering. Optionally, a second filter can be enabled by the means of this function. This filter allows the selection of destination or source address checking and group filtering. Only one address can be configured and it's respective mask for group filtering. The mask is 6 bits long and is used for comparing each of the MAC Address bytes. When set high, the MAC does not compare the corresponding byte of received destination address or source address with the configured MAC address. You can filter a group of addresses by masking one or more bytes of the address. When configuring the source address inverse filter, all the traffic that matches the configured source address will be rejected and the traffic that does not match the configured source address is accepted. The source address filter (either normal or inverse) result and the destination address filter result are AND'ed to decide whether the packet to be accepted. This means that the packet is dropped if either filter fails. The packet is accepted only if the packet passes both filters.

Parameters

in	*instance*	Instance number
in	*macAddr*	The physical address
in	*mask*	Group filter mask
in	*addrType*	Selects between destination address filter, source address filter or source address inverse filter
in	*enable*	If true, the application will receive all the traffic matching the selected filter; if false, it stops forwarding this kind of traffic.

7.2.6.34 Gmac_Ip_EnableVlan()

```
void Gmac_Ip_EnableVlan (
            uint8 Instance,
            const Gmac_Ip_VlanConfigType * VlanConfig )
```

Enable and set VLAN control for transmitted and received packets.

Parameters

in	*instance*	Instance number
in	*vlanConfig*	VLAN configuration structure

7.2.6.35 Gmac_Ip_SetTxOuterVlanTagForInsertion()

```
Gmac_Ip_StatusType Gmac_Ip_SetTxOuterVlanTagForInsertion (
            uint8 Instance,
            uint8 Ring,
            Gmac_Ip_VlanType VlanType,
            uint16 VlanTag )
```

Sets outer VLAN type and tag to be inserted by a particular transmission ring.

Parameters

in	*instance*	Instance number
in	*ring*	Ring number
in	*vlanType*	VLAN type (S-VLAN or C-VLAN) to be used
in	*vlanTag*	Value for the VLAN tag to be used

Return values

GMAC_STATUS_SUCCESS	The ring was successfully configured.
GMAC_STATUS_BUSY	The resource is busy and the ring could not be configured.
GMAC_STATUS_TIMEOUT	The ring configuration could not complete before expiration of timeout.
GMAC_STATUS_UNSUPPORTED	The function was called when VLAN replacement option was configured.

7.2.6.36 Gmac_Ip_SetTxOuterVlanTagForReplacement()

```
Gmac_Ip_StatusType Gmac_Ip_SetTxOuterVlanTagForReplacement (
            uint8 Instance,
            Gmac_Ip_VlanType VlanType,
            uint16 VlanTag )
```

Sets outer VLAN tag to be replaced for all transmission rings.

Parameters

in	*instance*	Instance number
in	*vlanType*	VLAN type (S-VLAN or C-VLAN) to be used
in	*vlanTag*	Value for the VLAN tag to be used

Return values

GMAC_STATUS_SUCCESS	The rings were successfully configured.
GMAC_STATUS_UNSUPPORTED	The function was called when VLAN insertion option was configured.

7.2.6.37 Gmac_Ip_SetTxInnerVlanTag()

```
void Gmac_Ip_SetTxInnerVlanTag (
            uint8 Instance,
```

```
        Gmac_Ip_VlanType VlanType,
        uint16 VlanTag )
```

Sets inner VLAN type and tag to be inserted/replaced on transmission.

Parameters

in	*instance*	Instance number
in	*vlanType*	VLAN type (S-VLAN or C-VLAN) to be used
in	*vlanTag*	Value for the VLAN tag to be used

7.2.6.38 Gmac_Ip_AddVlanTagToHashTable()

```
void Gmac_Ip_AddVlanTagToHashTable (
            uint8 Instance,
            uint16 VlanTag )
```

Adds a VLAN Tag to the Hash Table filter.

Parameters

in	*instance*	Instance number
in	*vlanTag*	Vlan Tag to be added to the hash table

7.2.6.39 Gmac_Ip_RemoveVlanTagFromHashTable()

```
void Gmac_Ip_RemoveVlanTagFromHashTable (
            uint8 Instance,
            uint16 VlanTag )
```

Removes a VLAN Tag from the hash table filter.

Parameters

in	*instance*	Instance number
in	*vlanTag*	Vlan Tag to be removed from the hash table

7.2.6.40 Gmac_Ip_InitSysTime()

```
void Gmac_Ip_InitSysTime (
            uint8 Instance,
            const Gmac_Ip_SysTimeConfigType * SysTimeConfig )
```

Initialize system time.

Parameters

in	*instance*	Instance number
in	*sysTimeConfig*	Pointer to a structure representing the configuration of the system time

7.2.6.41 Gmac_Ip_SetSysTimeCorr()

```
Gmac_Ip_StatusType Gmac_Ip_SetSysTimeCorr (
            uint8 Instance,
            Gmac_Ip_SysTimeCorrOffsetType Offset,
            uint32 SecondsUpdate,
            uint32 NanoSecondsUpdate )
```

Set system time correction.

Parameters

in	*instance*	Instance number
in	*offset*	Chooses between negative and positive correction
in	*secondsUpdate*	The seconds part of the update
in	*nanosecondsUpdate*	The nanoseconds part of the update

Return values

GMAC_STATUS_SUCCESS	The correction was set with no error.
GMAC_STATUS_TIMEOUT	The correction could not be set before expiration of timeout.

7.2.6.42 Gmac_Ip_SetRateRatioCorr()

```
Gmac_Ip_StatusType Gmac_Ip_SetRateRatioCorr (
            uint8 Instance,
            float64 RateRatio )
```

Set system time correction.

Parameters

in	*instance*	Instance number
in	*rateRatio*	Rate ratio is used to sync

Return values

GMAC_STATUS_SUCCESS	The correction was set with no error.
GMAC_STATUS_TIMEOUT	The correction could not be set before expiration of timeout.

7.2.6.43 Gmac_Ip_GetSysTime()

```
void Gmac_Ip_GetSysTime (
            uint8 Instance,
            Gmac_Ip_TimestampType * Timestamp )
```

Gets the current system time.

Parameters

in	*instance*	Instance number
out	*timestamp*	Pointer to a structure representing the current system time

7.2.6.44 Gmac_Ip_SetTxSchedAlgo()

```
void Gmac_Ip_SetTxSchedAlgo (
            uint8 Instance,
            Gmac_Ip_TxSchedAlgoType SchedAlgo )
```

Sets the transmitter scheduling algorithm.

Parameters

in	*instance*	Instance number
in	*scheme*	Transmitter scheduling algorithm.

7.2.6.45 Gmac_Ip_SetTxRingWeight()

```
void Gmac_Ip_SetTxRingWeight (
            uint8 Instance,
            uint8 Ring,
            uint32 Weight )
```

Sets the weight (in WRR scheduling algorithm) for a particular Tx ring.

Note: If multiple rings are configured and the WWR scheduling algorithm is selected, the weight must be programmed with a non-zero value. The maximum value for the weight is 0x64.

Parameters

in	*instance*	Instance number
in	*ring*	Tx ring
in	*weight*	Tx ring weight

7.2.6.46 Gmac_Ip_EnableTxStoreAndForward()

```
void Gmac_Ip_EnableTxStoreAndForward (
            uint8 Instance,
            uint8 Ring )
```

Enables the store and forward feature on the transmit path.

When the store and forward feature is enabled, the MAC starts to transmit data only when a complete packet is stored in the Transmit Queue.

Parameters

in	*instance*	Instance number
in	*ring*	Tx ring

7.2.6.47 Gmac_Ip_SetTxThreshold()

```
void Gmac_Ip_SetTxThreshold (
            uint8 Instance,
            uint8 Ring,
            Gmac_Ip_TxThresholdType ThresholdValue )
```

Sets transmit threshold levels.

Parameters

in	*instance*	Instance number
in	*ring*	Tx ring
in	*thresholdValue*	The threshold value

Chapter 8

Data Structure Documentation

8.1 Gmac_Ip_ManagementInfo Struct Reference

Management Frame information.

`#include <Gmac_Ip_Hw_Access.h>`

8.1.1 Detailed Description

Management Frame information.

Definition at line 109 of file Gmac_Ip_Hw_Access.h.

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